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Deep reinforcement learning-based optimization for machining process: Cutting parameters dynamic optimization considering tool condition degradation

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HIGHLIGHTS

- Tool wear and breakage monitoring with a multi-task teacher-student model.
- We reformulate cutting parameter optimization as an MDP, decomposing it into low-dimensional subprocesses.
- We employ a PPO algorithm with dual strategy heads for tool change and parameter adjustment.
- We evaluated our method on aero-engine casings to validate its effectiveness.

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Surface roughness
Production efficiency

ABSTRACT

In modern manufacturing, optimizing cutting parameters is crucial for enhancing efficiency and quality, yet dynamic tool condition degradation during machining process presents significant challenges to the cutting parameters optimization. Current research often neglects the tool condition degradation when optimizing cutting parameters, treating it as a static problem, limiting adaptability and real-time performance. To overcome these limitations, this article proposes a reinforcement learning (RL)-based dynamic optimization method for cutting parameters that incorporates tool condition degradation. Firstly, a multi-task learning teacher-student architecture is developed to comprehensively monitor tool condition using limited labeled data, which can provide accurate tool condition information for the subsequent cutting parameter optimization. Then the dynamic, high-dimensional, and coupled cutting parameter optimization problem is formulated as a Markov decision process (MDP). Finally, considering that the decision variables in this paper include a mixed action space of tool change and cutting parameter adjustment, we design a Proximal Policy Optimization (PPO)-based optimization algorithm. In this approach, the agent selects process parameters or decides whether to change tool for each cutting pass based on real-time machining state, thereby achieving efficient and high-quality manufacturing. Experimental validation was conducted on the actual manufacturing site of the areo-engine casing. The results show that the proposed tool condition monitoring (TCM) method can achieve a monitoring performance of R2 0.9484 for tool wear value and an f1-score of 0.8975 for breakage condition. Compared to ten multi-objective optimization algorithms, the proposed approach improves the optimization effect and computational efficiency by an average of 19.7185 % and 99.2615 %, respectively.

Nomenclature

VBi	Tool wear at ith cutting pass
B_i	Tool breakage state at *i*th cutting pass
x^i	Tool change operation at *i*th cutting pass
f_r^i	Feed rate at *i*th cutting pass (mm/r)
f	Feed per minute (mm/min)
n	Spindle speed (rpm)
v_c^i	Cutting speed at ith cutting pass (m/min)

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(*continued*)

a_p^i	Depth of cut at *i*th cutting pass (mm)
κ	Machining allowances (mm)
T_p	Total production time (s)
T_{rapid}	Rapid feed time (s)
T_{cut}	Cutting time (s)
T_{re}	Retracting time (s)
T_{tc}	Tool change time (s)
R_a	Surface roughness (μm)
R_{amax}^r	surface roughness in roughing stage (μm)
D	Initial diameter of workpiece (mm)
L_{cut}	Cutting length (mm)
MRR	Material remove rate (mm^3/min)
M_c	Tool material
$\gamma(^\circ)$	Rake angle of tool
$\kappa_r(^\circ)$	Principal cutting edge angle of tool
$\alpha(^\circ)$	Flank angle of tool
r_ε(mm)	Cutting edge radius
$\lambda_s(^\circ)$	Cutting edge Inclination
τ	Scaling factor
TCM	Tool condition monitoring
MTL	Multi-task learning
DRL	Deep Reinforcement learning
PPO	Proximal policy optimization
STTS	Single-task teacher-student architecture
MTTS-TCM	multi-task teacher-student architecture for TCM
DRL-TCDDO	The abbreviation of the proposed method in this paper
KL	Kullback-Leibler divergence p^s, p^t The output of student model and teacher model
$p^{tg}(\cdot)$	self-generated hard labels
$\alpha_s(t)$	Weight coefficient of two loss term
c_v	Coefficient of variation
MDP	Markov decision process
S	Set containing the state of each step s_t
A	Set containing the action of each step a_t
R	Reward function
P	Transition kernel
γ	Discount factor of reward
$\pi(\cdot\|\cdot)$	Policy
$HV(\cdot)$	Hypervolume volume calculation function
$GD(\cdot)$, $IGD(\cdot)$	Generational distance and, inverted generational distance calculation function
$SP(\cdot)$	Spacing indicator calculation function
r_s, r_m	Sparse reward and immediate reward
Λ	Lebesgue measure
υ	Reference point
PF^*	True Pareto frontier
$N(\cdot)$	Normalization operator
φ_1, φ_2	Balance factor of r_s and r_m
$V_\omega^{\pi_\theta}(s)$	State value function
$A^{\pi_\theta}(s,a)$	Advantage function
δ_t	Temporal difference
λ	Coefficient weight making a trade-off between Monte Carlo estimation and temporal difference
lr_{actor}, lr_{critic}	Learning rate of policy network and critical network
$r_t(\theta)$	Probability ratio

1. Introduction

The pursuit of high-quality and high-efficiency production has always been an important objective in the aerospace and other high-end manufacturing industries. In the cutting process, how to configure the appropriate cutting parameters is a crucial factor to improve production efficiency and processing quality [1]. As the direct executor in cutting operations, the performance of the cutting tool serves as a vital basis for the selection of cutting parameters and is also a key factor influencing the quality of machining and productivity [2]. However, most of the current cutting parameter optimization researches ignore the tool condition, including: tool wear and breakage, resulting in sub-optimal results [3,4]. Therefore, it is urgent to consider the tool condition degradation, dynamically optimize cutting parameters and control tool replacement.

During the cutting process, the tool condition inevitably deteriorates, mainly manifested as the increase in the flank wear value of cutting tools. However, since the processing characteristics such as large removal amount of aerospace parts and difficult processing of materials, the tool condition will deteriorate rapidly during the cutting process, accompanied by the chipping and breakage, in which the tool breakage will have a more serious impact on the workpiece quality, leading to catastrophic consequences [2]. Besides, the machining allowance tends to be much larger than the allowable depth of cut, which requires multi-pass turning, comprising multi-pass roughing and a single-pass finishing [5]. Thus, the initial set cutting parameters are difficult to adapt to the rapid degradation of tool condition, so that the machining process is not carried out in an optimal state, which may cause failure of the tool during cutting, affecting the processing quality and efficiency, which is be shown in Fig. 1. we can observe from figure that if the cutting parameters are adjusted according to the tool condition before the i + 1 pass or the tool is replaced immediately, the remaining useful life (RUL) of the tool can be extended, which enables the cutting tool to perform the cutting operation in an excellent condition.

Although several studies have indicated that tool wear represents a crucial determinant in the selection of cutting parameters, few scholars have considered the dynamic changing tool condition in the optimization process of cutting parameters [6,7], and most of the studies introducing the tool condition only assume that the tool wear remains fixed or single-pass cutting throughout the cutting process. For example, Tian et al. regarded different wear conditions as an additional decision variable to realize the energy-efficient turning operations [7]. Zhang et al. introduced tool wear in the cutting force model to develop an improved energy consumption model for milling operations [8]. Xie et al. utilized NSGA-III to optimize energy consumption and surface quality considering influence of tool wear at a certain material removal rate [9]. Additionally, almost all studies do not consider tool chipping or breakage in the optimization process, in which tool breakage can alter the effective geometry shape of a cutting tool like built-up-edge (BUE), affecting its cutting performance, with a more serious impact on machining quality and may directly lead to tool failure [10]. Wojciechowski et al. pointed out that BUE will form during the cutting process regardless of whether coated tools are used, affecting the effective cutting geometry and performance of the tool [11]. Fig. 2 demonstrates that when the tool occurs chipping, we can adjust the cutting parameters to overcome the decline in tool performance. However, if the chipping progresses to breakage, immediate tool replacement is imperative, and continue cutting will cause processing quality problems. Besides, most scholars still treat cutting parameter optimization problems that consider tool condition as static optimization problems applying metaheuristic algorithm, in which the length of each solution is pre-set and fixed [4]. In the actual machining process, the tool condition is dynamic and real-time change, the length of the final solution depends on the tool condition and the cutting parameters selected each cutting pass, which suggests that conventional metaheuristic approaches struggle to address scenarios where the length of decision variables fluctuates, owing to their static decision space [12]. Although the published methods have made some contributions to the optimization of cutting parameters, they still have the following limitations:

- A strong interdependence exists between machining parameters and tool condition. Specifically, the configured cutting conditions directly influence the progression of tool degradation, for instance, as the feed rate increases, the friction coefficient trends upwards; employing advanced cooling/lubrication technology such as MQL + CuNPs can significantly reduce tool wear [13]. While the current condition of the cutting tool, in turn, determines the optimal choice of machining parameters [14]. However, the majority of existing methods ignore this dependency by either fail to incorporate the tool condition into the optimization process or only optimize for a given level of tool wear. In addition, almost all literatures ignore the impact of tool chipping and breakage on the selection of cutting parameters during machining.

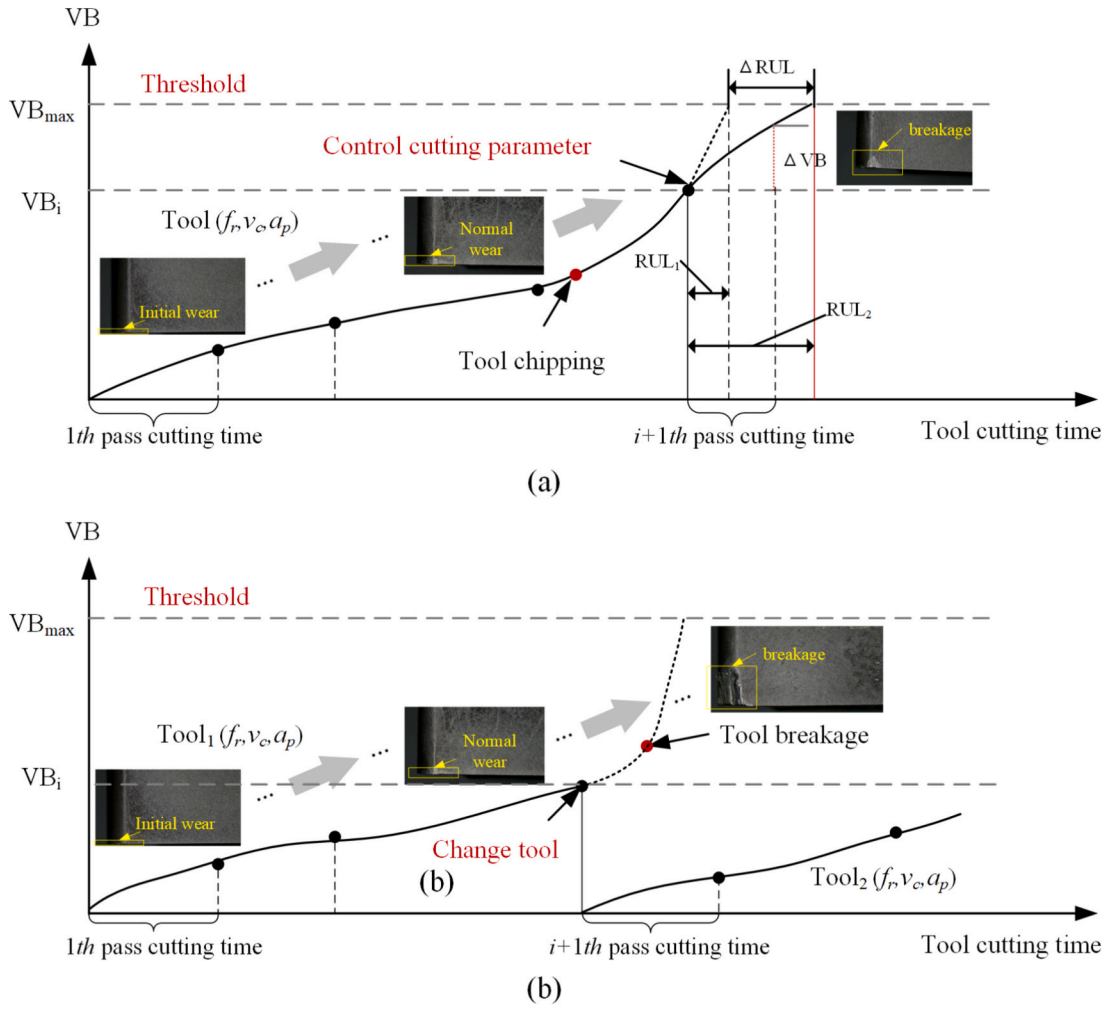

Fig. 1. Tool condition degradation in multi-pass cutting: (a) control cutting parameter, (b) tool replacement.

- During cutting process, the tool condition is inevitably degraded, and cutting parameters need to be updated in a short period of time [15]. While traditional methods are usually only suitable for fixed machining configurations (i.e., fixed workpieces, machining allowance and tool condition), when machining configurations have changed, the traditional methods need to re-run the optimization experiments to solve the optimal cutting parameters, which is often time-consuming and costly, typically taking several hours or even days [16]. This makes them ill-suited to adapt to the rapidly changing tool conditions.
- Under a fixed machining allowance, the number of cutting passes is influenced by the depth of cut. Nevertheless, given the fixed decision space of metaheuristic optimization methods, the traditional method usually predetermines the number of cutting passes. Subsequently, the cutting parameters for each pass are optimized independently, without accounting for the interdependence between the cutting parameters and the number of passes [17]. This approach restricts the diversity of potential solutions.

To solve the aforementioned issues and fill the research gap, this paper proposes a novel dynamic cutting parameter optimization method based on deep reinforcement learning, considering comprehensive tool condition degradation (DRL-TCDDO) for high-quality and high-efficiency turning operations. Reinforcement learning (RL) has been demonstrated to exhibit remarkable efficacy in different optimal control problems [18], which learns manufacturing process knowledge by interacting with the machining environment [19]. Besides, unlike traditional meta-heuristic methods, RL does not set up a fixed decision space in advance, but transforms optimal control into a sequential decision-making process [20], and replaces the objective function by long-term average reward, so that the model can consider the influence on the subsequent cutting parameters when selecting the current cutting parameters. While deep learning technology has strong nonlinear fitting ability and obvious advantages when dealing with high-dimensional data [21]. Hence, integrating DL and RL to explore heuristic strategies, the complex dynamic optimization problem of cutting parameters in machining can be effectively addressed [22]. Considering that accurately monitoring the tool status is the basis for achieving dynamic optimization of cutting parameters, this paper first proposes a semi-supervised tool condition monitoring (TCM) model, leveraging multi-task learning (MTL) and pseudo-label technology. Subsequently, the cutting parameter optimization problem is formulated as a Markov decision process (MDP) and a virtual machining environment is then constructed according to the current actual machining configuration and the TCM model, to facilitate the training of an agent. Given that the decision variables encompass both cutting parameter adjustments and tool change decisions, which constitute a mixed action space with both discrete and continuous actions, we introduce the proximal policy optimization (PPO) algorithm to train agent. The main contributions of this paper are as follow:

- We establish a novel multi-task teacher-student architecture to accurately monitor the process of tool condition degradation, in which teacher-student architecture can handle a large amount of

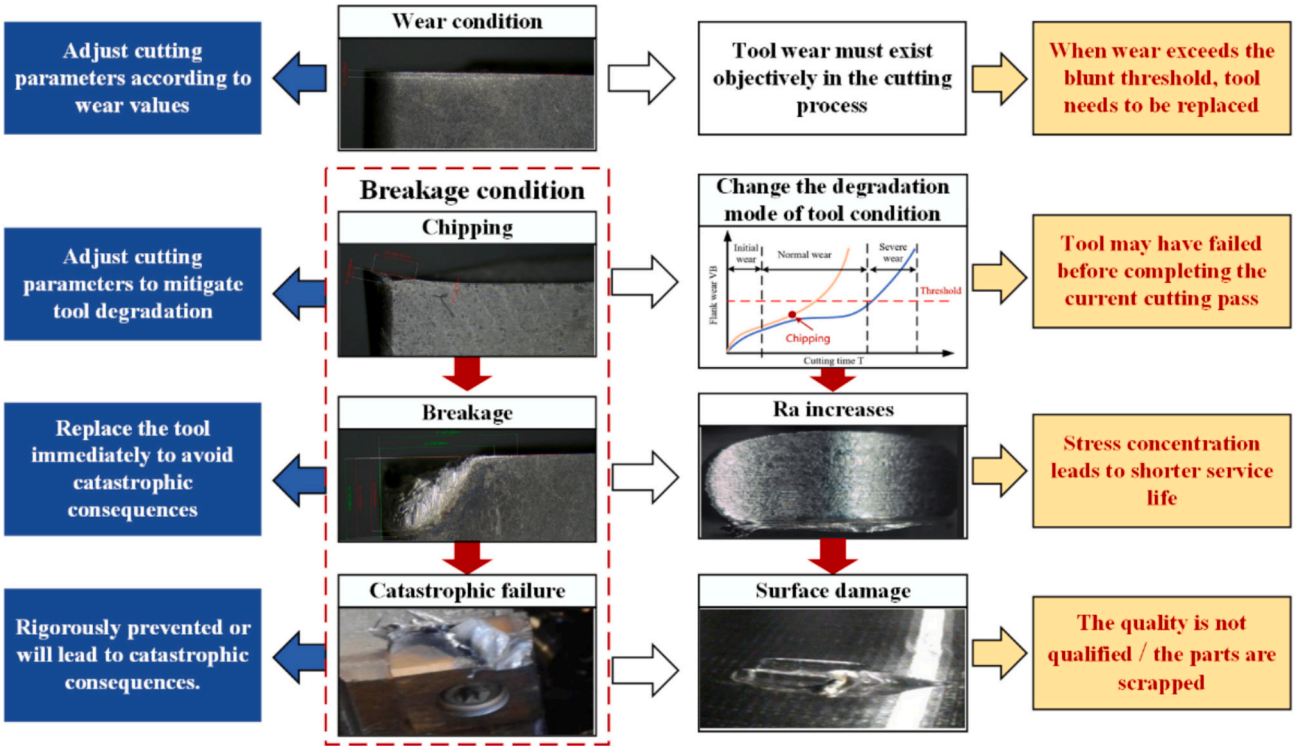

Fig. 2. Influence of different tool conditions on machining process.

unlabeled data generated in the manufacturing process and assign corresponding pseudo-labels. While MTL can extract the common features of tool wear and breakage, which indicates that the proposed TCM method can simultaneously monitor the tool wear value and breakage state with a small amount of labeled data.

- A novel cutting parameter optimization method of multi-pass turning operation is proposed, which can adapt to the rapidly changing tool condition. The optimization problem is first transformed into MDP, which converts the original high-dimensional optimization problem into several low-dimensional subordinate decision processes and then a virtual machining environment is constructed according to the proposed TCM model and the domain knowledge of the machining process.
- We employ PPO technique to develop a RL algorithm, which is trained in an actor-critic (AC) architecture. Therein, the actor network outputs corresponding actions into the established virtual machining environment to obtain reward values, which are sparse rewards composed of multi-objective optimization indicators and instant rewards composed of tool condition. Then, the critic network evaluates the current states of the virtual machining environment to guide the actor network's strategy training.
- We have executed systematic and comprehensive experiments in a real aero-engine casing machining site and obtained corresponding experimental data to verify the advantages and feasibility of our proposed method. The experimental results show that DRL-TCDDO can achieve accurate and efficient decision-making of machining process parameters under continuous degradation of tool conditions, thereby improving quality and efficiency.

The remainder of this paper is organized as follow: Section 2 comprehensively reviews the current literature in TCM and cutting parameter optimization. Section 3 presents the problem statement. Section 4 describes the proposed semi-supervised TCM method. Section 5 describes the proposed cutting parameter optimization approach. Section 6 executes a comprehensive case study using the real aero-engine casing machining data, and followed by a summary and future works in Section 7.

2. Related work

2.1. Tool condition monitoring

Accurate and comprehensive TCM is the basis of dynamic cutting parameter optimization in this paper, which has emerged as a research hotspot recently. Although direct monitoring methods have high accuracy, they always need to interrupt the machining operation, and the visual sensor has demanding requirements on the machining environment, making it difficult to meet the real-time requirements of the machining site. With the rapid development of technologies such as deep learning algorithms and the Internet of Things (IoT), data-driven methods are more applicable to modern manufacturing [23–25]. Li et al. [26]presented a novel audio-based tool wear monitoring model, which combines the dependent component analysis and ensemble learning strategy to accurately monitor the wear state. Sun et al. [27] proposed a semi-supervised learning framework based on auto-encoder to monitor the tool breakage. Huang et al. [28] applied the probabilistic neural network to realize the real-time tool breakage detection. Duan et al. [29] conducted real-time prediction of tool wear values by combining Siamese neural network (SNN) and support vector regression (SVR), where SNN provides rich feature information for SVR's final prediction of tool wear. Considering that tool condition degradation is irreversible and has typical timing characteristics, many studies have introduced recurrent neural networks, transformer and their variants to extract degradation information to realize TCM [25,30,31].

However, the reliability of the aforementioned research relies heavily on extensive labeled datasets for model training. In the actual machining process, acquiring large amounts of labeled data is often impractical, as it requires interrupting cutting operations and consumes significant time costs [32,33]. This implies that real data possesses an imbalance, scarce label samples, and a noisy nature, fortunately, there is a large amount of unlabeled data in engineering practice, but how to effectively use these large amounts of unlabeled data and extract the

information of the machining process is ignored by most data-driven methods currently [2]. Furthermore, most data-driven approaches monitor tool wear and breakage as two separate tasks, ignoring their inherent correlations at the data level. This may lead to reduced model performance when monitoring tool condition degradation cycles involving both phenomena. Fig. 3 illustrates the feature changes of acoustic emission (AE) signals in the time-frequency domain when the tool is breakage during cutting. It can be seen that when the tool condition degradation process is accompanied by wear and breakage, the signal features will interfere with each other.

2.2. Cutting parameters optimization methods

Cutting parameter optimization in turning operations requires precise modelling of the correlations between machining parameters and performance objectives, including production efficiency and workpiece quality. Some studies explore optimal cutting parameters by designing a series of machining experiments (DoE), which regard cutting parameters as parametric variables and measure the related metrics to establish the statistical curve-fitting functions [34]. Oscar et al. [35] combined the response surface methodology (RSM) with the desirability function to acquire the quantitative relationship between cutting parameters and responses. Philip Selvaraj et al. [14] utilized the Taguchi method to design the turning experiment and applied the analysis of variance (ANOVA) to explore the effects of cutting speed and feed rate on surface roughness, cutting force and tool wear. Although these methods are simple and practical, they require a large amount of experimentation as a basis, with long optimization cycles and high costs. Moreover, due to limited experimental conditions, the solution space is discretized, usually resulting in locally optimal solutions [36].

Cutting parameter optimization is a highly nonlinear problem with multiple constraints and objectives. Compared with DoE methods, metaheuristic algorithms are widely used because of their good global searching ability. Yildiz et al. [37] proposed an optimization algorithm based on artificial bee colony to optimize the cutting parameters in multi-pass turning process. Leo Kumar et al. [38] utilized the genetic algorithm (GA) to optimize parameters with minimum processing time and minimum surface roughness (Ra) as objectives. Bennett et al. [39] compared different metaheuristic algorithms, such as GA, particle swarm optimization (PSO) algorithm and simulated annealing (SA) algorithm, when optimizing the surface roughness of turning, and the results showed that the cutting parameters of GA optimization obtained the minimum surface roughness. These approaches can quickly approach the optimal solution in the single-pass cutting. In multi-pass cutting, the selection of the current cutting parameters will lead to changes in the tool condition of all subsequent passes, and the number of passes will also be affected. The decision dimensions of metaheuristic methods are fixed, making it difficult to capture the coupling relationships between decision variables. Besides, metaheuristic approaches often only address static optimization problems; they must be re-solved whenever the machining state changes, which takes a significant amount of time [40].

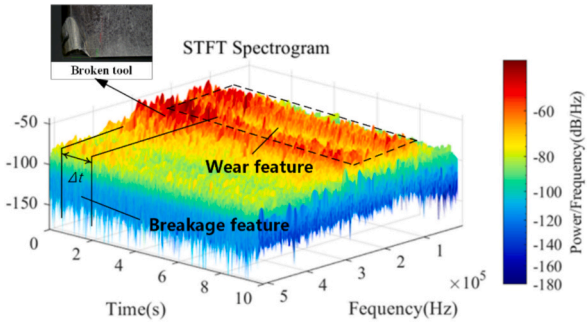

Fig. 3. STFT of AE signals.

In recent years, RL has been widely used in robotics [41], dynamic scheduling [42] and process optimization [43] due to its strong self-learning ability. Unlike the blind and random violent global search of metaheuristic algorithms [44], RL learns manufacturing domain knowledge by interacting with the machining environment to maximize long-term returns, and appropriately explores unknown strategies in the learning process to avoid falling into local optimality. In addition, when the machining state changes, such as tool condition, the cutting parameters suitable for the current machining state can be quickly obtained according to the machining knowledge that has been learned by RL. These characteristics of RL make it possible to dynamically optimize cutting parameters under rapid change of tool condition.

2.3. Cutting parameters optimization model considering tool condition degradation

As mentioned above, the tool condition has a great influence on the selection of cutting parameters, and some literature have introduced tool wear into the optimization of cutting parameters. Liu et al. [45] utilized digital twin technology to achieve accurate surface roughness prediction and adaptive optimization of process parameters, incorporating only the tool wear state as an additional variable for surface roughness prediction into the optimization process. Tian et al. [8] proposed a cutting parameter optimization method considering tool wear in order to reduce energy consumption and achieve low-carbon manufacturing. Zhang et al. [6] combined off-line estimation with real-time monitoring to propose a milling parameter optimization with cutting efficiency as the objective, and monitored the tool wear state by the spindle current during the machining process, and then optimized milling parameters in real time according to constraints. Xiao et al. [36] optimized the cutting parameters of multi-pass turning operations by integrating the advantages of meta-learning and RL, taking the tool wear state as a constraint and characterizing it as whether the surface roughness after the current cutting exceeds the constraint. Zhang et al. incorporated the effects of stochastic tool wear into an energy consumption optimization model for optimization of cutting parameters in micro-milling, where tool wear progression is obtained by monitoring cutting forces. However, most literature currently focuses only on the optimization of cutting parameters under the assumption that the tool condition remains constant during a single machining process or that tool wear increases in a fixed incremental manner, which does not conform to the actual machining conditions. Furthermore, to our best knowledge, almost all literature has not considered the occurrence of tool breakage in the optimization process, which poses a greater threat to machining quality and may directly lead to tool failure, part scrap, and other issues.

2.4. Summary of comparison with previous cutting parameter optimization methods

As mentioned above, DoE methods are limited by high experimental costs and a discretized solution space often leading to local optima, while metaheuristic algorithms, despite their global search capability, struggle with the dynamic and coupled nature of multi-pass turning operations, requiring costly re-optimization for any change in machining state. Due to these limitations, in recent years some advanced artificial intelligence methods based on RL have been applied to the dynamic optimization of cutting parameters. Wang et al. [46]. presented a DRL-based pipeline to enhance the production efficiency and quality in manufacturing, applying the double deep Q network (DDQN) to solve the milling parameter optimization. Li et al. [47]. optimized the cutting parameters in the milling process by considering cutting energy consumption and machining cost as objectives, using the proposed BP-TD3 method, where BP is used for energy consumption modelling and TD3 is used to solve the cutting parameter optimization problem. Besides, there

are also studies that have constructed virtual processing environments based on graph attention networks, utilized multi-agent systems to optimize processing parameters, and introduced evolutionary learning to enhance the stability of multi-agent training [48]. Wang et al. [49]. proposed a novel multi-task DRL method for machining parameter optimization in batch processing systems is proposed, enabling the collaborative consideration of quality and efficiency. Lu et al. [50]. proposed an energy-saving integrated optimization of cutting parameters and tool paths based on hierarchical reinforcement learning (HRL). The upper layer plans the cutting parameters for each pass on a macro time scale, while the lower layer, on a micro time scale, uses the planned cutting parameters to execute multiple tool path extensions and provides feedback to the upper layer. Furthermore, the research has found that the cutting tool condition has a significant impact on the effectiveness of the optimization of cutting parameters. Therefore, some scholars have taken the condition of the cutting tool into account during the process of optimizing cutting parameters [4,15,51,52]. Some articles have also conducted research on the interpretability of the processes involved in manufacturing process optimization, in order to prevent the optimization results from confusing the decision-making of the operators [53–56]. However, the introduction of explainability will to some extent reduce the optimization effect and increase the computational complexity [57], thereby affecting the real-time performance of the model. In addition, current explainable artificial intelligence (AI) methods have not yet been explored in the optimization of cutting parameters and tool change decision-making in turning operations. Table 1 summarizes the current literature on cutting parameter optimization based on AI methods.

As depicted in the table above, prior studies have made significant strides in applying AI for cutting parameter optimization. However, a critical gap remains in the integration of TCM within a real-time dynamic optimization framework. Although tool conditions are one of the important factors affecting the selection of cutting parameters, a considerable portion of the above literature has either ignored them or only taken tool wear into consideration, assuming that the tools would not experience more severe situations such as chipping or breakage. Moreover, they can hardly handle a mixed action space (a combination of continuous parameter adjustments and discrete tool change decisions), which aligns more similar to real-world manufacturing decision-making.

Table 1
Comparison of the proposed method with current AI-based approaches for cutting parameter optimization.

Methods (years)	Optimization objective	Tool wear condition	Tool breakage condition	Action space
Wang et al. [46] (2022)	Quality and efficiency	No	No	Discrete
Li et al. [47] (2022)	Energy and cost	No	No	Continuous
Li et al. [48] (2023)	Quality and efficiency	No	No	Continuous
Wang et al. [49] (2025)	Quality and efficiency	No	No	Continuous
Lu et al. [50] (2025)	Energy and efficiency	No	No	Continuous
Xiao et al. [36] (2021)	Energy, cost and efficiency	No	No	Continuous
Zhao et al. [52] (2024)	Energy, cost and efficiency	Yes, but assume constant in multiple workpieces	No	Continuous
Li et al. [15] (2022)	Energy and efficiency	Yes, but assume constant in a workpiece	No	Continuous
Gao et al. [51] (2025)	Energy, cost and efficiency	Yes, but assume constant in a workpiece	No	Continuous
Xiao et al. [4] (2023)	Energy and efficiency	Yes	No	Mix (Continuous and Discrete)
Ours	Quality and efficiency	Yes	Yes	Mix (Continuous and Discrete)

Therefore, in response to the above issues and the limitations of current research, this paper proposes a novel dynamic cutting parameter optimization method called DRL-TCDDO, and the specific implementation and experimental validation will be described in detail in the subsequent sections.

3. Problem statement of cutting parameter optimization

The turning operation is a highly prevalent and pivotal manufacturing process that finds extensive application across diverse industries [58], in which the workpiece is securely clamped within a chuck, rotated around its axis, and a non-rotating single-point cutting tool is employed to remove the undesired material, thereby fabricating axisymmetric parts [59]. During this process, the tool condition gradually deteriorates with increasing wear through three stages of wear: initial wear, normal wear and severe wear. However, due to the processing characteristics of aerospace parts, the tool may experience chipping or breakage during the cutting process, resulting in a sudden change in the tool wear law or even direct failure. The decline in tool condition can lead to an increase in cutting force and a deterioration in the surface quality of the parts. If the cutting parameters are fixed, it will further aggravate the degradation of tool condition. in the processing field, cutting parameters are usually adjusted based on manual experience, but this adjustment method is fixed and conservative, making it difficult to balance both processing quality and efficiency [46]. Hence, we first collect the AE monitoring signals and process information of the machining process to monitor the tool condition, and then determine the current cutting parameter combination based on the current tool condition to improve production efficiency and processing quality.

3.1. Optimization objective

Multi-pass turning operation includes multi-pass roughing and a single-pass finishing. During the roughing stage, a large amount of material needs to be removed, and larger cutting parameter combinations are often selected to improve efficiency. However, excessively large cutting parameters can easily lead to tool wear and breakage failure, causing surface defects and increasing tool change time. The finishing stage focuses on processing quality, often opting for smaller cutting parameter combinations. The two stages are inherently interconnected, as the cutting parameters chosen in the rough machining stage will affect the results of the finishing stage [60]. For example, when a larger cutting depth is chosen in the roughing stage, the allowance for finishing will decrease accordingly. Therefore, to balance the objectives of the roughing and finishing stages, this paper studies the multi-objective optimization problem of quality and efficiency. Considering that surface roughness is a key indicator of processing quality, and in aerospace parts, more than 80 % of fatigue cracks are caused by unsatisfactory surface roughness, this paper chooses surface roughness to measure processing quality [61]. The two objectives are described as follows:

$$\min F(X) = \left(\min T_p, \min R_a\right) \tag{1}$$

where T_p and R_a denote total production time and surface roughness, respectively. X is the decision variables that will be introduced later.

3.2. Decision variable

It should be noted that the optimization of cutting parameters in this paper includes two aspects:1. If the tool currently monitored does not exceed the blunt threshold and no damage occurs, the cutting parameters are adjusted; 2. If the model cannot give a reasonable combination of cutting parameters to avoid the failure of the cutting tool, it will directly give the instruction to change the tool. Therefore, the decision variables X can be defined as:

$$X = \left\{ \left(x^i, f_r^i, v_c^i, a_p^i | VB_i, B_i\right) \ldots, \left(x^{N_r}, f_r^{N_r}, v_c^{N_r}, a_p^{N_r} | VB_{N_r}, B_{N_r}\right), \left(x^f, f_r^f, v_c^f, a_p^f | VB_f, B_f\right) \right\} \tag{2}$$

where $f_r^i, v_c^i, a_p^i$and x^i are tool feed rate (mm/r), cutting speed (m/min), depth of cut (mm) and tool change action of the ith cutting pass, respectively. Superscript f represents the finishing pass. VB_i and B_i represent the tool wear and breakage condition of the ith cutting pass.

3.3. Production time

The production time T_p (s) reflects the processing efficiency, which is crucial for enterprises. This paper divides the T_p into 4 parts, including rapid feed time, cutting time, retracting time and tool change time. As can be seen from Fig. 4, before starting the cutting process, the tool will first return to the machine reference point (point A) to wait for the CNC program to start the machining instruction; The tool is then quickly fed to the start of the cutting position (point B); Then start cutting from point B to point C and then rise to point D and quickly retract from point D to E; Finally, the tool moves to the path of the second cutting and starts a new cycle, and when the machining allowance returns to zero, the tool returns to the reference point again to prepare for the machining of the next workpiece. Thus, T_p can be calculated as follows:

$$T_p = T_{rapid} + T_{cut} + T_{re} + T_{tc} \tag{3}$$

where T_{rapid}, T_{cut}, T_{re} and T_{tc} represent the rapid feed time, cutting time, retracting time and tool change time, respectively. T_{rapid} can be calculated as follows:

$$T_{rapid} = \frac{L}{\frac{1000 f_{rapid}}{60}} = \frac{3L}{50 f_{rapid}} \tag{4}$$

where L (mm) is the distance from point A along the path of the red dotted line to point B. f_{rapid} (m/min) is the rapid feed speed of the machine tool. The T_{cut} can be derived by the following formula:

$$n = \frac{1000 v_c}{\pi D} \tag{5}$$

$$f = \frac{n f_r}{60} \tag{6}$$

$$t_{cut}^i = \frac{L_{cut}}{f} \tag{7}$$

$$T_{cut} = \sum_{i=1}^{N_r+1} t_{cut}^i \tag{8}$$

where D (mm) is the current diameter of the workpiece. n (rpm) is the spindle speed, and f (mm/min) is the feed per minute. t_{cut}^i and L_{cut} indicate the time of ith cutting pass and cutting length, respectively. Tre is the time for the tool to return to the cutting starting point after each cutting, which can be obtained from the following formula:

$$T_{re} = \sum_{i=1}^{M} t_{re}^i \tag{9}$$

$$t_{re}^i = \frac{L_{re}}{\frac{1000 f_{rapid}}{60}} = \frac{3 L_{re}}{50 f_{rapid}} \tag{10}$$

where t_{re}^i is the time of rapid retracting of the ith pass. L_{re} is the length of the tool from point D along the red dotted line to the start point of the next cut.

3.4. Constraint analysis

In the machining process, to ensure the performance of machine tools and cutting tools, meet the processing safety and workpiece processing requirements, it is necessary to restrict the variables related to

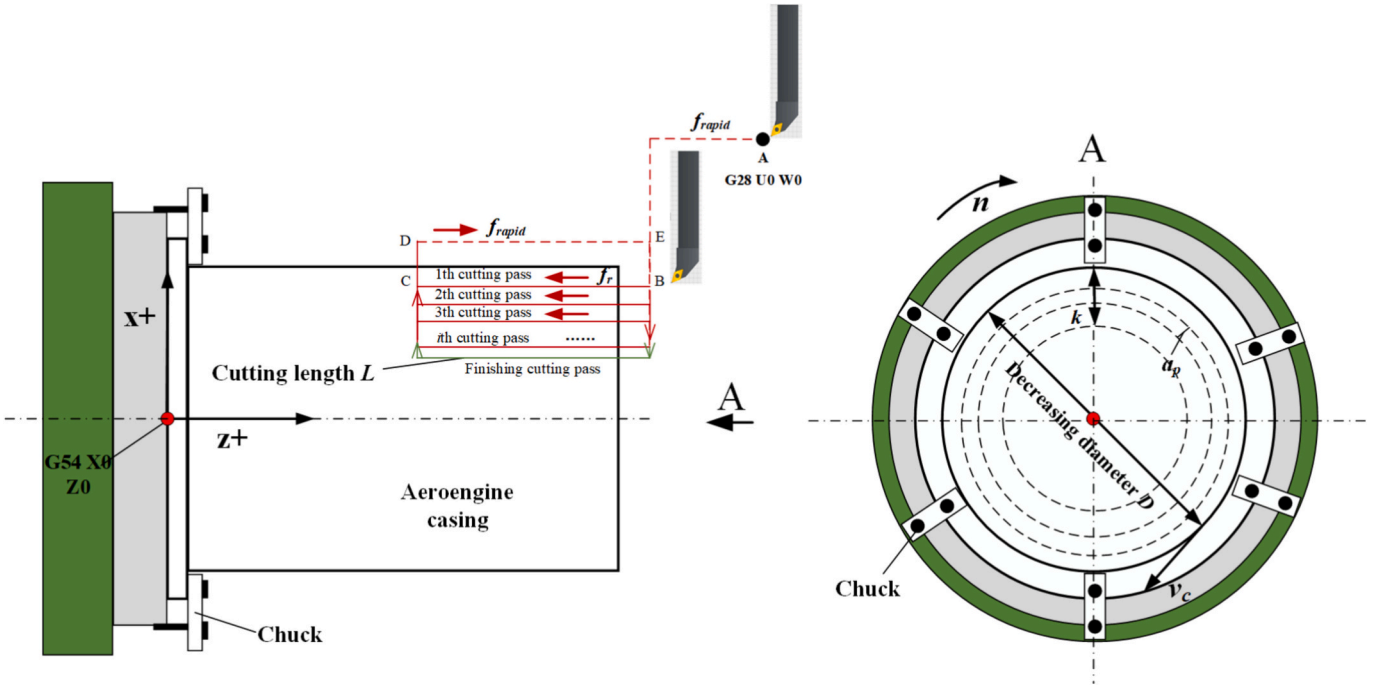

Fig. 4. Schematic diagram of the cutting path for machining the outer cylindrical feature by turning process.

the optimization of cutting parameters, which can be summarized as follows:

$$v_{cmin} \le v_c \le v_{cmax} \tag{11}$$

$$f_{rmin} \le f_r \le f_{rmax} \tag{12}$$

$$a_{pmin} \le a_p \le a_{pmax} \tag{13}$$

$$x = \begin{cases} 1, & if\ change\ tool \\ 0, & else \end{cases} \tag{14}$$

$$\kappa = \sum_{i=1}^{N_r} a_p^i + a_p^{N_r+1} \tag{15}$$

$$B_i \le B_{\max}^r, B_f \le B_{\max}^f \tag{16}$$

$$VB_i \le VB_{\max}^r, VB_f \le VB_{\max}^f \tag{17}$$

$$R_a{}^r \le R_{amax}^r \tag{18}$$

where Eq. (11)–(13) ensure that the selected cutting parameters are within a reasonable safe range and Eq. (14) shows that the tool change is a discrete variable of either 0 or 1. Eq. (15) ensures the cutting continuity, that is, the depth of cut of rough machining and finishing adds up to the machining allowance. Eq. (16) and Eq. (17) indicate that the tool wear value should not exceed the corresponding threshold value and should be avoided from breakage, where $B_{\max}^r$and $B_{\max}^f$ denote the tool breakage states acceptable during the roughing and finishing phases, respectively. $VB_{\max}^r$ and $VB_{\max}^f$denote the tool wear values acceptable during the roughing and finishing phases, respectively. Eq. (18) avoids surface anomalies caused by extreme cutting conditions in the roughing stage.

Considering that surface roughness is not only related to process parameters, but also affected by more process information such as tool condition, cutting time, workpiece material and machining method. Instead of trying to deduce the display formula between surface roughness and various variables according to the complex cutting principle, it is better to directly use artificial intelligence algorithms to establish the mapping relationship between various process information and R_a [62,63]. The details of the surface roughness modelling process are presented in Section 4.2.

4. TCM and surface roughness modelling

4.1. A multi-task teacher-student architecture for TCM

In this section, we give a detailed introduction to a novel multi-task teacher-student architecture for TCM (MTTS-TCM). The overall framework of the proposed method is shown in Fig. 5. The collected historical data is first divided into labeled data and unlabeled data according to weather there is a tool condition label for model training. Then we introduce a hard parameter sharing MTL architecture to construct teacher model and student model, of which the structures are the same, but their training procedures are different. Next, the well-trained teacher model assigns pseudo-labels to the unlabeled data, and after filtering the low-confidence pseudo-labels, they are used for student model training to provide more effective information for them. Finally, the student model is trained using the high-quality pseudo-labels provided by the teacher model and a small amount of labeled data.

The MTL model we proposed generally has a shared layer and two task-specific layers, in which the shared layer can extract common features between wear and breakage tasks, and the task-specific layer can extract features unique to each task. The shared layers first take the processed AE signals as input, and then use a convolutional neural network (CNN), which is widely used for feature extraction, to extract the tool condition degradation information, where the CNN comprises an input layer, convolutional layers, and pooling layers. The computational process of CNN is shown in Eq. (19):

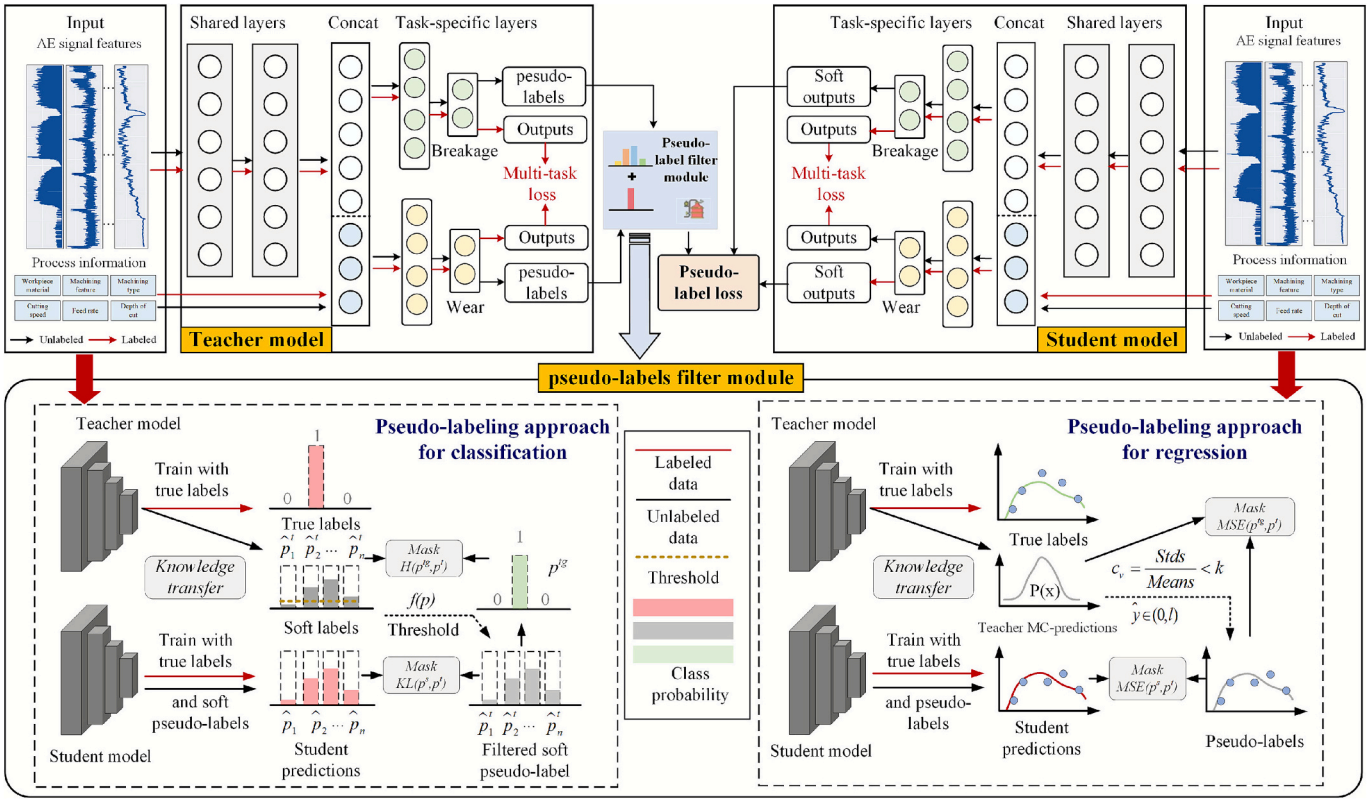

Fig. 5. The overall framework of MTTS-TCM.

$$M_i^l = f\left(\sum_j M_j^{l-1} \otimes w_j^l + b_i^l\right) \tag{19}$$

where M_i^l is the ith feature vector of the lth layer extracted by ith convolutional kernel. w_j^l and b_i^l denote the trainable convolution kernel parameters and bias parameters, respectively. $\otimes$ is the convolution operator, and $f(\cdot)$ denotes the activation function. Considering that process information can provide prior knowledge related to processing conditions, this paper will concatenate the features extracted from the shared layer with the process information to form a hybrid feature vector. The feature vector is subsequently fed into two task-specific layers which both consist of three fully connected layers.

Given that a large amount of unlabeled data in engineering practice has not been effectively utilized, we have constructed a teacher-student structure for TCM model training. First, the teacher model uses labeled data to train for T1 epochs in a supervised learning way, and then begins to assign pseudo-labels to the unlabeled data. Following this, the student model is trained using the generated pseudo-labels and labeled data. Given the limited quantity of labeled data, training the teacher model solely on the labeled data may result in overfitting, thereby diminishing the quality of the pseudo-labels. Consequently, after T1 training epoch, the teacher model is updated by incorporating both the pseudo-label data and the labeled data. And this process continues consistently in the subsequent training cycles, that is, the teacher model not only provides pseudo-labels to the student model but also updates itself simultaneously. It is important to emphasize that we do not derive hard pseudo-labels for student model training directly from the maximum probability of the teacher model output, as is common in traditional pseudo-labeling. Instead, we implement an improved softmax output layer that transforms the outputs of the teacher model into a probability distribution, which can be represented as follows:

$$p_i^t = \frac{exp(z_i/\tau)}{\sum_j exp(z_i/\tau)} \tag{20}$$

where z_i and p_i^t represent the logits for each class and the predicted probability distribution, respectively. The scaling factor τcontrols the smoothness of the probability distribution. When it set to 1, the above formulation degenerates to standard softmax. This probabilistic output is what we term soft labels. Unlike hard labels, these soft targets preserve richer information about class relationships, particularly when the probability distribution exhibits high entropy, thereby maintaining discriminative information about negative classes. For example, an excessively wear sample is closer to the distribution of a breakage sample than that of a normally wear sample. To assess the discrepancy between the soft pseudo-labels produced by the teacher model and the output distribution of the student model, we introduce the Kullback-Leibler (KL) divergence, which is defined as follows:

$$D_{KL}(p^s \| p^t) = \sum_{x \sim D} p^s \log \frac{p^s}{p^t} \tag{21}$$

where $p^s$and p^t represent the output of the student model and teacher model, respectively. By minimizing the KL divergence, the student model can effectively learn from the information derived from the unlabeled data provided by the teacher model. Besides, the student model also requires labeled data to train, therefore, the loss function of the student model consists of pseudo-label term and labeled data term, which can be written as follows:

$$\begin{aligned} L_s &= \alpha_s(t) D_{KL}(p^s \| p^t) + (1-\alpha_s(t)) H(p^g, p^s) \\ &= \alpha_s(t) \sum_{i=1}^{n} p^s\left(y|x_i^u\right) \log \frac{p^s\left(y|x_i^u\right)}{p^t\left(y|x_i^u\right)} + (1-\alpha_s(t)) \left(-\sum_{i=1}^{n} p^g\left(y|x_i^l\right) \log\left(p^s\left(y|x_i^l\right)\right) \right) \end{aligned} \tag{22}$$

where x_i^u and x_i^l are the unlabeled data and labeled data, respectively, which will be used for teacher and student training. $p^g(\cdot)$ and n represent the ground truth distributions and the class numbers, respectively. $\alpha_s(t)$ is a weight coefficient to balance the importance of two loss terms, which will increase adaptively as the number of training epoch increases, which can be written as follows:

$$\alpha_s(t) = \begin{cases} 0 & t < T_1 \\ \dfrac{t - T_1}{T_2 - T_1} \alpha_{s-max} & T_1 < t < T_2 \\ \alpha_{s-max} & t > T_2 \end{cases} \tag{23}$$

where t denotes the current training epoch; α_{s-max}, T_1, and T_2 are hyperparameters set to 1, 10, and 300 in this research. Furthermore, we give the loss function form of the teacher model:

$$\begin{aligned} L_t &= H(p^g, p^t) + \alpha_t(t) H(p^{tg}, p^t) \\ &= \left(-\sum_{i=1}^{n} p^g\left(x_i^l\right) \log\left(p^t\left(x_i^l\right)\right) \right) + \alpha_t(t) \left(-\sum_{i=1}^{n} p^{tg}\left(x_i^u\right) \log\left(p^t\left(x_i^u\right)\right) \right) \end{aligned} \tag{24}$$

Unlike the student model, the pseudo-label loss of the teacher model is calculated by self-generated hard labels$p^{tg}(\cdot)$. $\alpha_t(t)$is a weight coefficient to balance the importance of two loss terms. This indicates that the pseudo-labels generated by the teacher model are not fixed, but are continuously improved as the student model and the teacher model are trained alternately. By iteratively re-evaluating historical low-confidence pseudo-labels, it avoids permanently discarding them and thus prevents the loss of key information (such as early tool chipping).

Considering that low-quality pseudo-labels will interfere with model training and limit model performance, this paper introduces a simple and efficient filtering mechanism to screen high-confidence pseudo-labels for student model training. Specifically, we exclusively preserve the pseudo-labels that exhibit a maximum probability exceeding the established threshold. This criterion implies that the teacher model possesses a high degree of confidence in these labels for the unlabeled data, which can be mathematically expressed as follows:

$$\frac{1}{B^u} \sum_{b=1}^{B^u} \mathbf{1}(max(p(y|x^u) \geq \gamma)) L^u \tag{25}$$

where $\mathbf{1}(\cdot) \in \{0, 1\}$ is an indicator function that assumes a value of 1 if and only if the specified input conditions are satisfied. $p(y|x^u)$ is the soft pseudo-label generated by the teacher model for unlabeled data x^u. B^u and L^u represent the batch size and loss function of unlabeled data, respectively. However, the above filtering method cannot be applied to regression tasks which outputs an exact value rather than a probability distribution. In view of this, this paper uses Monte Carlo dropout and prior knowledge to filter pseudo-labels for regression tasks, which can be defined as follows:

$$\frac{1}{B^u} \sum_{b=1}^{B^u} \mathbf{1}\left(c_v = \frac{\sigma}{\mu} \leq k\right) \mathbf{1}(\widehat{y}^t \in (0, l)) L^u \tag{26}$$

where $\widehat{y}^t$ and c_v are the regression pseudo-labels generated by the teacher model and the coefficient of variation (CV), respectively. μ and σ are the mean value and standard deviation value of the T predictive pseudo-labels (generated by the teacher model) by Monte Carlo dropout, respectively. The CV value directly reflects the model's uncertainty in predicting this sample, which can be used to filter out the vast majority of high-noise, high-uncertainty samples. Ultimately, only those samples that are 'not too uncertain for the model but have relatively stable predicted values' are retained. l represents half of the maximum geometric length that is perpendicular to the direction of the cutting edge of the tool. The above pseudo-label learning process is shown in Algorithm

1.

Algorithm 1. Pseudo-label learning with teacher-student architecture.

Input: labeled data $D^l = \{(x_j^l, y_j^l)\}_{j=1}^{N_l}$ and unlabeled data $D^u = \{(x_j^u)\}_{j=1}^{N_u}$
Output: well-trained student model S with parameter θ_s
Initialize teacher model T with parameter θ_t, student model S with parameter θ_s, τ, $\gamma^{'}$, k, Tl, E, T, l
WHILE epoch < E+1 **DO**:
  // Teacher model training on limited labeled data
  FOR each batch (x^l, y^l) in D^l **DO**:
    $\hat{y}^l = T(x^l, \theta_t)$
    Using softmax($\hat{y}^l$)and y^l to calculate loss and update θ_t
  END FOR
  // Pseudo-label generation and filtering
  IF epoch $\geqslant Tl$ **DO**:
    FOR each batch x^u in D^u **DO**:
      $\hat{y}^u = T(x^u, \theta_t)$
      // Filter high-confidence pseudo-labels
      IF max(softmax($\hat{y}^u$))> $\gamma^{'}$ **DO**: // For breakage task with max probability threshold
        y_0^p = softmax($\hat{y}^u / \tau$)
      END IF
      FOR m=1 to T **DO**:
        $y_{VB}^{(T)} = T(x^u, \theta_t$, dropout=True) // For wear task with Monte Carlo dropout
      END FOR
      μ_{vb}=mean($y_{VB}^{(T)}$), σ_{vb}=std($y_{VB}^{(T)}$),CV=σ_{vb}/μ_{vb}
      IF CV<k and $\mu_{VB} \in (0, l)$ **DO**:
        $y_{VB}^p = \mu_{VB}$
      END IF
    END FOR
  END IF
  // Student model training with combined data
  FOR each combined batch (x^l, y^l) and (x^u, y^p) DO:
    $\hat{y}_s^l = S(x^l, \theta_s)$, $\hat{y}_s^u = S(x^u, \theta_s)$
    Calculate pseudo-label loss based on y^p and $\hat{y}_s^u$ using KL divergence KL($\hat{y}_s^u \| y^p$) (MSE for wear task: MSE($\hat{y}_s^u, y^p$))
    Calculate labeled data loss based on y^l and $\hat{y}_s^l$ using cross entropy H($\hat{y}_s^l, y^l$) (MSE for wear task: MSE($\hat{y}_s^l, y^l$))
    Update θ_s
  END
  // Teacher model update pseudo-labels with self-generated hard pseudo-labels and limited labeled data
  FOR each combined batch (x^l, y^l) and (x^u, y^p) DO:
    $\hat{y}_T^l = T(x^l, \theta_t)$, $\hat{y}_p^u = T(x^u, \theta_s)$, here the pseudo-labels are hard labels
    Update teacher model parameter θ_t based on Eq. (23)
  END
END WHILE

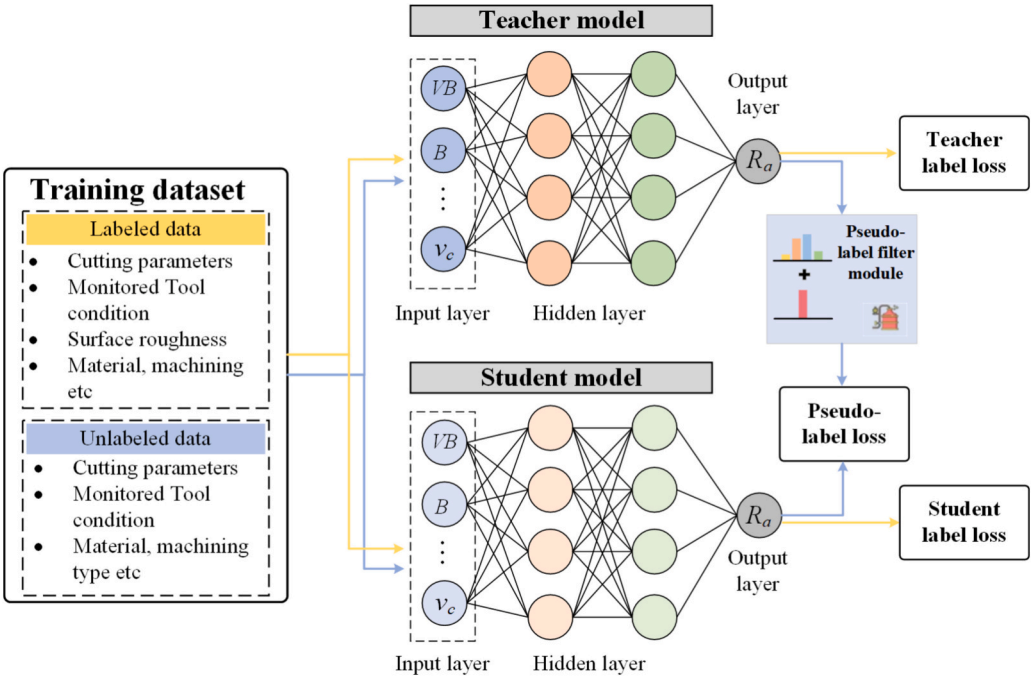

Fig. 6. Surface roughness prediction model based on teacher-student architecture.

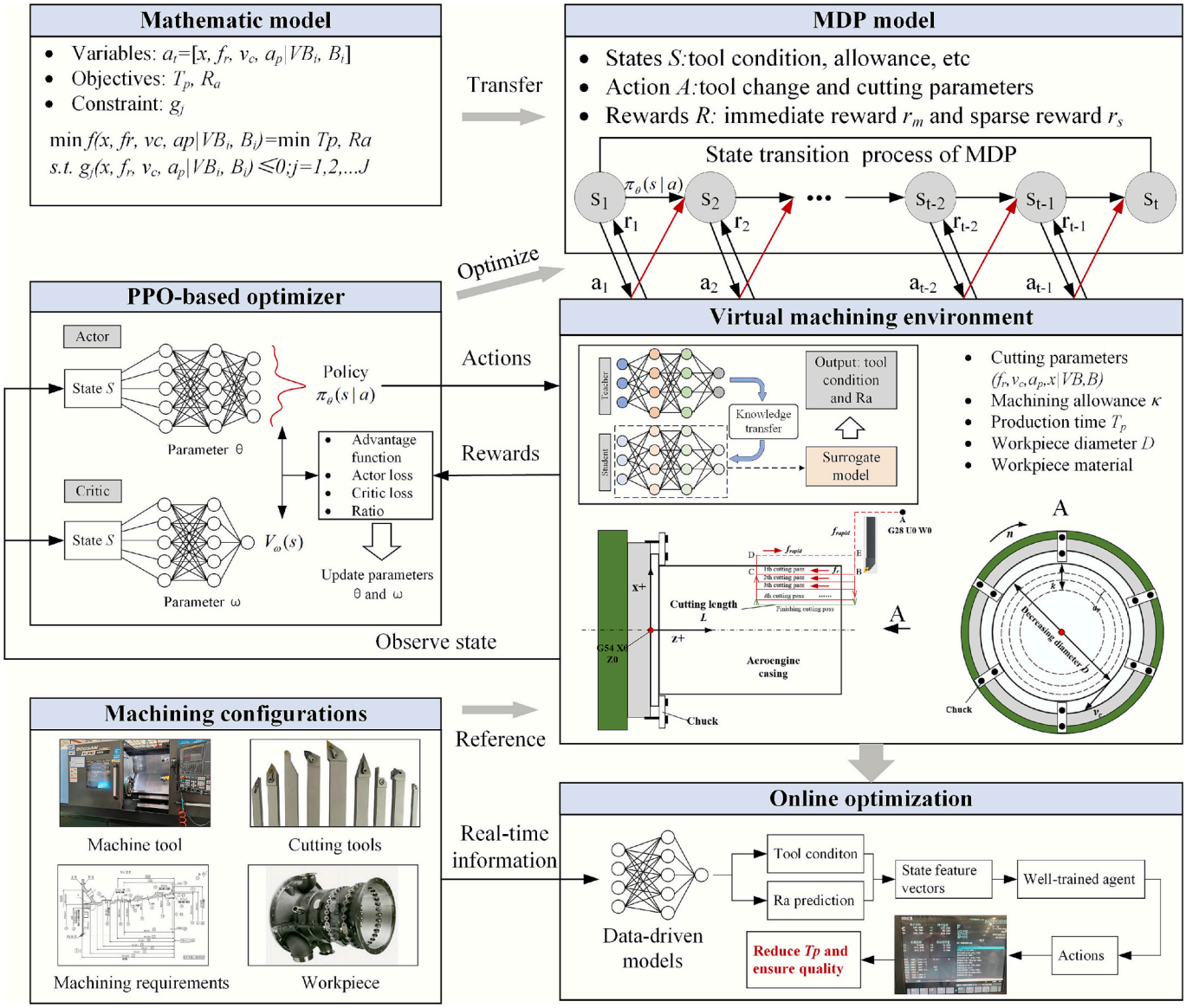

Fig. 7. The overall framework of DRL-TCDDO.

4.2. Surface roughness modelling

As mentioned in section 3.1, surface roughness is one of the important factors affecting the fatigue behavior and useful life of workpiece, and is affected by several variables such as cutting parameters, machining methods and tool condition. Therefore, under the premise of fully considering the tool condition, this paper utilizes deep learning and pseudo-label techniques to estimate the surface roughness.

The surface roughness evaluation method framework is shown in Fig. 6. The proposed MTTS-TCM is firstly used to obtain the current tool condition and take it as the input variable of the surface roughness prediction model, where the input variables also include: cutting parameters, workpiece material, cutting time, machining type and machining feature. Following a similar operation as in Section 4.1, the student model is then trained on high-quality pseudo-labels provided by the teacher model and limited labeled data. Finally, the well-trained student model is applied to evaluate the surface roughness.

5. PPO-based dynamic cutting parameter optimization framework

In this section, a PPO-based cutting parameter decision-making method is developed to optimize cutting parameters of multi-pass turning operations under dynamic tool condition, aiming at improving the processing quality and production efficiency. The proposed DRL-TCDDO framework is illustrated in Fig. 7 which mainly includes two stages: offline training stage and online optimization stage. In the offline training stage, the dynamic cutting parameter optimization problem is first transformed to MDP. Then, PPO algorithm design and virtual machining environment construction are carried out respectively. Finally, the agent interacts with the virtual machining environment continuously to accumulate the knowledge of cutting parameter decision. In the online optimization stage, the proposed MTTS-TCM monitors the tool condition through collected real-time AE data and process parameters to determine the current tool wear value and breakage state. The well-trained agent then selects cutting parameters or makes a tool change decision based on the tool condition and the current machining state.

5.1. MDP formulation

To solve the optimization problem in this paper using RL method, the cutting parameter control problem must be transformed into MDP. During each run of multi-pass turning, the agent firstly observes the state

of the current processing environment $s_t \in S$, then uses policy $\pi(a_t|s_t)$ to generate an action $a_t \in A$ to input into the virtual environment. The environment then updates the machining states $s_{t+1} \in S$ based on the selected action and gives a corresponding reward $r_t \in R$. The agent optimizes its strategy based on the reward given by the environment and performs the next action, repeating the process until the end of the current part processing. The MDP process is described as MDP = {S,A,P, R,γ} and defined as follows:

1) **State space** $S = \left\{ s_i = \left[VB_i, B_i, D_i, L_i, \kappa_i, MRR_i, R_a^i, x_{i-1}, f_r^{i-1}, v_c^{i-1}, a_p^{i-1} \right] \right\}$ contains a series of states s_i, each state s_i includes the information of ith cutting pass that can describe the machining environment. VB_i and B_i denote tool wear value and breakage state, respectively. D_i and L_i denote workpiece diameter and cutting length. κ_i is the current machining allowance. $x_{i-1}, f_r^{i-1}, v_c^{i-1}$ and a_p^{i-1} represent the selected actions of previous cutting pass. R_a^i is the surface roughness. MRR_i denotes the Material removal rate, which is an important indicator of processing efficiency, and is defined as follows:

$$MRR = v_c \cdot a_p \cdot f_r \tag{27}$$

2) **Action space** $A = \left\{ a_i = \left[x_i, f_r^i, v_c^i, a_p^i \right] \right\}$ contains a series of actions a_i, each a_i includes whether to change the tool x_i and corresponding cutting parameters, which are selected for each cutting pass according to the state space s_i.

3)**Transition kernel** $P : S \times A \times S \rightarrow [0,1]$ indicates the transition probability $P(s_{i+1}|s_i, a_i)$ that the environment moves to the next state s_{i+1} after action a_i is performed in state s_i.

4) **Reward** $R : S \times A \rightarrow R$ is the reward function that receive from the virtual environment, which is the core element designed to guide the learning of the agent according to the objective function. The design of the reward function directly affects the learning efficiency and the final performance of the agent. Given that machining quality and production efficiency are two conflicting goals, to effectively evaluate the advantages of the current strategy, this paper introduces two multi-objective optimization evaluation indicators: hypervolume volume (HV) and Generational distance (GD) to construct a reward function, which can be defined as follows:

$$HV(X, v) = \Lambda\left(\bigcup_{\in X} \left[T_p^i, v_1 \right] \times \left[R_a^i, v_2 \right] \right) \tag{28}$$

$$GD(X, PF^*) = \frac{\sqrt{\sum_{x \in X^*} \left(\min_{p^* \in PF^*} \left\| x - p^* \right\| \right)^2}}{|X|} \tag{29}$$

where $X = \{T_p, R_a\}$ is the objective value of nondominated solution set, $v = \{v_1, v_2\}$ represents the reference point. Λ is the Lebesgue measure used for calculating the hypervolume of X and the reference point v. PF^* is the true Pareto frontier. $\|\cdot\|$ denotes the Euclidean distances used for calculating the distance from each solution in the solution set to the true front, and $|X|$ is the number of solution set. Since it is difficult to obtain the true Pareto frontier in actual problems, we manually select excellent solution sets to approximate PF^* based on domain knowledge and historical data. The bigger HV and smaller GD of the X indicates the better approximation and distribution of the PF^*. Thus, the reward function can be written as follows:

$$r_s = HV(X, v) - GD(X, PF^*) \tag{30}$$

However, the evaluation of total production time and surface roughness is only conducted when the machining features of the current workpiece are finished, which indicates that r_s is hard to provide feedback on the actions selected for each cutting pass of agent. Therefore, we introduce the Eq. (16)–(18) and MRR to construct immediate reward, which is provided immediately after finishing each cutting pass, and it can be defined as follows:

$$r_m = r_{VB} + r_B + r_{R_a} + N(MRR) \tag{31}$$

where r_{VB}, r_B and r_{Ra} represent the reward values for the tool wear, tool breakage condition and surface roughness of the current cutting pass, respectively. If the corresponding pre-set threshold is not exceeded, then r_{VB}, r_B and r_{Ra} are all set to 1, otherwise, they are set to −1. $N(\cdot)$ represents the normalization operator. Therefore, the final reward function can be described as follows:

$$\begin{aligned} r &= \varphi_1 r_h + \varphi_2 r_m \\ &= \varphi_1 (HV(X, v) - GD(X, PF^*)) + \varphi_2 (r_{VB} + r_B + r_{R_a} + N(MRR)) \end{aligned} \tag{32}$$

where φ_1 and φ_2 are the balance factor of the sparse reward r_h and the immediate reward r_m.

5) **Discount factor** $\gamma \in [0,1]$ determines that the weight of current rewards and future rewards and directly affects the long-term decision strategy of the agent.

5.2. Virtual environment description

In RL, agent training is carried out by interacting with the environment and conducting corresponding decision experiments to learn strategies. However, in the early stage of training, the agent will provide incorrect strategies, leading to catastrophic consequences in the real machining environment, such as machine tool damage and workpiece scrap. Therefore, it is very necessary to construct a virtual machining environment based on real machining configurations to replace the real machining environment as the trial-and-error object of the agent.

The virtual machining environment in this paper mainly includes four modules: state monitoring module, which collects AE signals and processes real-time machining status information like tool wear value (VB), breakage condition (B), workpiece diameter (D), cutting length (L), and machining allowance (κ); action execution module, which converts the agent's output actions (tool change decision x, feed rate f_r, cutting speed v_c, cutting depth a_p) into machining parameters; core Simulation Module, which obtains key indicators such as material removal rate (MRR), total production time (T_p), tool condition (VB and B) and surface roughness (R_a) based on manufacturing knowledge and data-driven models; reward calculation module, which generates immediate rewards and sparse rewards based on multi-objective optimization metrics. Through these modules, the virtual manufacturing environment first calculates the cutting time and MRR of the current cutting pass based on the actions provided by the agent. Then, according to the method proposed in Section 4.1, it updates the tool condition by using process information such as cutting parameters, machining features, machining methods, and cutting time. Next, update the surface roughness based on the method in Section 4.2. Finally, based on the designed reward function and the updated state, the virtual environment calculates the reward value. A detailed description of the virtual machining environment is given in Algorithm 2. Since this paper mainly focuses on the influence of cutting parameters and tool conditions on machining efficiency and quality, the constructed virtual machining environment differs from the real machining environment mainly in the following aspects: tool wear value, breakage state, and surface roughness are derived from trained data-driven models instead of multiple actual measurements; complex dynamics in real processing such as

machine tool vibration, thermal deformation, microscopic physical phenomena in the tool-workpiece contact area are ignored, as well as the coolant effects; real environmental interferences like ambient temperature and humidity changes, machine tool power supply fluctuations, workpiece material batch differences, and operator intervention impacts are not simulated.

Algorithm 2. Virtual environment of multi-pass turning operations.

Input: s_i, a_i
Output: s_{i+1}, r_i, info // info provides additional information about the environment
Initialize T_p, R_a, s
WHILE $\kappa_i > 0$ **DO**:
  IF $x_i = 1$ **DO**: // change tool
    Initialize VB_i, B_i
  END IF
  IF $\kappa_i - a_p^i < 0$ **DO**:// finishing stage
    $a_p^i = \kappa_i$
  END IF
  Update D_i based on a_p^i and machining feature type, take cylindrical turning as an example:
    $D_{i+1} = D_i - 2a_p^i$
  Update T_p^i and Calculate MRR_i using Eq. (3) and Eq. (26) $T_p^i \to T_p^{i+1}, MRR_i \to MRR_{i+1}$
  Update tool wear value VB_i and tool breakage state B_i according to process information such as: cutting parameters $\{f_r^i, v_c^i, a_p^i\}$, cutting time T_{cut} and workpiece material $VB_i \to VB_{i+1}, B_i \to B_{i+1}$
  Evaluate surface roughness R_a^{i+1} based on tool condition VB_{i+1}, B_{i+1} and process information.
  Update machining allowance $\kappa_{i+1} = \kappa_i - a_p^i$
  IF $\kappa_{i+1} == 0$ **DO**: // reward in finishing stage
    IF $VB_{i+1} > VB_{\max}^f$: $r_{VB} = -1$ **ELSE:** $r_{VB} = 1$
    END IF
    IF $B_{i+1} > B_{\max}^f$: $r_B = -1$ **ELSE:** $r_B = 1$
    END IF
    IF $R_a^{i+1} > R_{a\max}^r$: $r_{R_a} = -1$ **ELSE:** $r_{R_a} = 1$
    END IF
    Calculate immediate reward r_m using Eq. (30) and evaluate T_p and R_a
    Calculate sparse reward r_s using Eq. (29)
    Calculate total reward based on Eq. (31)
  ELSE: // reward in roughing stage
    IF $VB_{i+1} > VB_{\max}^r$: $r_{VB} = -1$ **ELSE:** $r_{VB} = 1$
    END IF
    IF $B_{i+1} > B_{\max}^r$: $r_B = -1$ **ELSE:** $r_B = 1$
    END IF
    Calculate immediate reward r_m using Eq. (30)
  Update state space $s_i \to s_{i+1}$
END WHILE

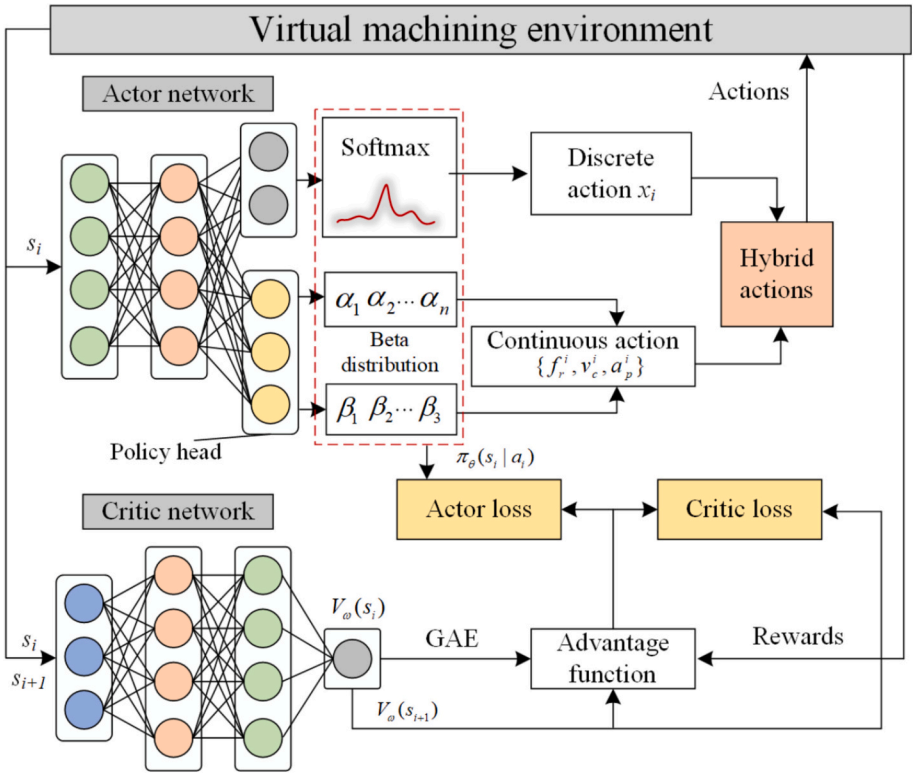

Fig. 8. The framework of PPO-based optimizer.

5.3. Optimizer designing and training

5.3.1. Construction of PPO-based optimizer

The action space in the optimization problem of this paper is a mixed action space where both continuous and discrete actions exist. Considering PPO algorithm can be applied to tasks including continuous action space and discrete action space, and it has the advantages of strong training stability and low computational complexity [64]. Therefore, we adopt PPO algorithm to build the optimizer, where the proposed optimizer is mainly composed of two networks: actor network and critic network. The following part will provide a detailed introduction to the structure of the optimizer.

The actor network firstly takes the current machining state s as the input, which includes two fully connected (FC) layers, and two separate policy heads for generating discrete and continuous actions. The specific structure is shown in Fig. 8. For tool change action x_i, the actor network will first output probabilities provided by Softmax to establish a Categorical distribution, and then use it to sample discrete actions, which can be shown as follow:

$$\pi_{\theta_d}(x_n|s) = \frac{\exp(logits_n)}{\sum_n \exp(logits_n)}, n \in \{1,2\} \tag{33}$$

where θ_d are the parameters of actor network for discrete actions. $Logits_n$ denotes the output of the actor network. For cutting continuous action $\left\{f_r^i, v_c^i, a_p^i\right\}$, Gaussian distribution is usually used for sampling, but it is an unbounded distribution, which is usually necessary to truncate or force the action of limiting samples to be in the range of 0 to 1 after action sampling, introducing additional complexity and potential numerical problems. Considering that the Beta distribution has an output range of between 0 and 1, and its numerical calculation is relatively stable, especially when processing boundary values. Therefore, this paper applies Beta distribution to sample continuous actions, which can be shown as follows:

$$\pi_{\theta_c}(a_n|s) = Beta(\alpha_n, \beta_n), n \in \{1,2,3\} \tag{34}$$

where θ_c are the parameters of actor network for continuous actions. a_n represents the cutting parameter adjustment actions sampled by the policy. α_n and β_n represent the parameters of Beta distribution. Furthermore, the sampled continuous actions need to be converted to the actual action range before being input into the environment:

$$a_{actual} = a_{n\min} + a_n(a_{n\max} - a_{n\min}) \tag{35}$$

where $a_{n\max}$ and $a_{n\min}$denote the lower and upper bounds of three cutting parameters.

The critic network provides the corresponding state value $V_\omega^{\pi_\theta}(s)$ based on the current machining states to assist the actor network exploration of learning policy. We also use the FC layers to build the value network and thereby obtain the state value, as shown below:

$$V_\omega^{\pi_\theta}(s_t) = \mathrm{Tanh}(F_V(s_t)) \tag{36}$$

where $F_V(\cdot)$ and $\mathrm{Tanh}(\cdot)$ are respectively the critic network and its activation function. After obtaining the state value $V_\omega^{\pi_\theta}(s)$, we introduce the generalized dominance estimation (GAE) method to estimate the advantage function $A^{\pi_\theta}(s,a)$ which measures the advantage of choosing a certain action over other actions in a specific state, which can be formulated as follows:

$$A_t^{GAE(\lambda)} = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l} \tag{37}$$

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t) \tag{38}$$

where $\lambda \in [0,1]$ is an additional hyperparameter introduced in GAE used to make a trade-off between Monte Carlo estimation and temporal difference (TD) estimation to achieve a balance between reducing variance and reducing bias. $\delta_t$denotes the TD, r_t is the reward obtained after taking an action.

5.3.2. Training of PPO-based optimizer

To optimize the weights of actor network to find the optimal policy, this paper applies policy gradient approach to maximize the following objective:

$$J(\theta) = \mathbb{E}_{\pi_{\theta_{old}}}[\min(r_t(\theta)A^{\pi_{\theta_{old}}}(s,a), clip(r_t(\theta\quad), 1-\varepsilon, 1+\varepsilon)A^{\pi_{\theta_{old}}}(s,a))] \tag{39}$$

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)} \tag{40}$$

$$\theta \leftarrow \theta + lr_{actor} \cdot \nabla_\theta J(\theta) \tag{41}$$

where $\pi_\theta(a_t|s_t)$ and $\pi_{\theta_{old}}(a_t|s_t)$ denote the current policy and old policy, respectively. $r_t(\theta)$ is the probability ratio. Clip(r_t, 1-ε, 1 + ε) indicates the clipping function that restricts r_t within [1-ε, 1 + ε], which ensure the step size range of policy updates is reasonable. ε is a hyperparameter. lr_{actor} represents the learning rate of actor network. Considering that this paper requires the policy for optimizing both discrete actions and continuous actions simultaneously, we calculate the probability ratios of the discrete and continuous parts respectively, and combined the continuous strategy and the discrete strategy:

$$\log\pi_\theta(a|s) = \log\pi_{\theta_d}(a_d|s) + \log\pi_{\theta_c}(a_c|s) \tag{42}$$

$$r_t(\theta) = r_t^d(\theta) \cdot r_t^c(\theta) \tag{43}$$

where $\pi_{\theta_d}(a_d|s)$ and $\pi_{\theta_c}(a_c|s)$ represent the discrete policy and continuous policy, respectively. $r_t^d(\theta)$ and $r_t^c(\theta)$ respectively denote the probability ratio of discrete policy and continuous policy.

For critic network, of the goal is to learn the state value function by minimizing the mean square error, which is defined as follows:

$$L(\omega) = \frac{1}{2}\mathbb{E}\left[(r_t + \gamma V_\omega(s_{t+1}) - V_\omega(s_t))^2\right] \tag{44}$$

$$\omega \leftarrow \omega + lr_{critic} \cdot \nabla_\omega L(\omega) \tag{45}$$

where $V_\omega(s_t)$ and $V_\omega(s_{t+1})$ are the current state-value function and next state-value function, respectively. r_t represents the current rewards obtained and γ is the discount factor. lr_{critic} indicates the learning rate of critic network. Algorithm 3 lists the detailed procedure of the PPO-based optimizer.

Algorithm 3. PPO-based optimizer for dynamic cutting parameter optimization.

Input: lr_{actor}, lr_{critic}, γ, λ, virtual machining environment
Output: Optimal parameter θ of actor network
Initialize actor network parameter θ and critic network parameter ω using Orthogonal Initialization method
Initialize memory buffer M and its max size M_{max}
Initialize minibatch size B for training
FOR episode=1 to T **DO**:
  Reset virtual machining environment and obtain initial machining states s_t
  WHILE $\kappa_j > 0$ **DO**:
    Select discrete actions $a_t^d \sim \pi_{\theta_d}(a_d \mid s_t)$ and continuous actions $a_t^c \sim \pi_{\theta_c}(a_c \mid s_t)$
    Execute selected actions in the virtual machining environment
    Obtain rewards r_t and next state s_{t+1}
    Store transition (s_t, a_t^d, a_t^c, r_t, s_{t+1}) in M
    Update s_t to s_{t+1}
  END WHILE
  // Policy optimization
  IF M == M_{max} **DO**:
    // Traverse the trajectories in M
    FOR each minibatch transitions **DO**:
      Sample B transitions (s_t, a_t^d, a_t^c, r_t, s_{t+1}) from M
      Obtain TD errors based on Eq. (37)
      Calculate actor loss $J_a(\theta)$ and $J_c(\theta)$ according to Eq. (38)
      Update $J_a(\theta)$ and $J_c(\theta)$ using Adam optimizer based on Eq. (40)
      Calculate critic loss $L(\omega)$ using Eq. (43)
      Update $L(\omega)$ using Adam optimizer based on Eq. (44)
    END FOR
  END IF
END FOR

6. Case study

6.1. Experimental details

To realize dynamic optimization of cutting parameters, it is necessary to design a corresponding data acquisition scheme to provide effective data support for the proposed DRL-TCDDO training. Therefore, We conducted an experiment at a certain production site for aero-engine casing, where we collected data generated during the turning operations of certain several types of aero-engine casings, which is shown in Fig. 9.

Table 2
Mechanical properties of aero-engine casing workpiece.

Mechanical properties	ZTC4	K4169
Density (g/cm^3)	4.45	8.24
Tensile Strength (MPa)	895	1275
Yield strength (MPa)	825	1030
Elongation (%)	6	8
Modulus of Elasticity (GPa)	114	200
Rockwell hardness (HRC)	33	40

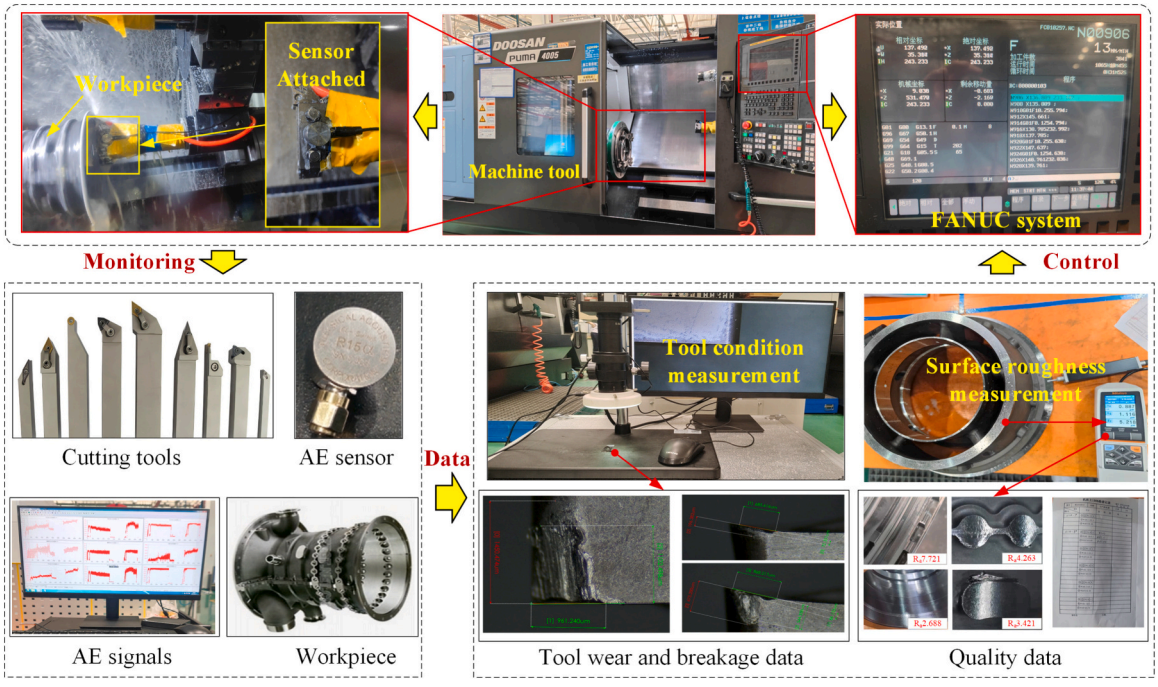

Fig. 9. Machining system and data acquisition platform.

Table 3
Main parameters of CNC machine tool.

Parameter	Value
X/Z axis travel range	352/1105 mm
Rapid feed of X axis	16 m/min
Rapid feed of Z axis	20 m/min
Maximum spindle speed	2000 rpm
Maximum turning length	1042 mm
Maximum turning diameter	Φ550mm
Maximum torque of spindle	2379NN·m

Table 4
Technical specifications and measurement settings of devices.

Device name	item	Value
AE sensor	Sampling frequency	1 MHz
	Preamp	40 dB
	Operating frequency range	50-450 kHz
	Sample length	1024
	Threshold	45 dB
	Peak frequency	146.48 kHz
Portable surface profiler	Measuring range	−200 μm ~ 160 μm
	Measuring speed	0.5 mm/s
	Measuring force	0.75 mN
	Sampling length	0.8 mm

The aero-engine casings involved in the machining process include compressor casings, power turbine casings and combustor chamber casings. Their materials are mainly ZTC4 titanium alloy and K4169 nickel-based superalloys, both are difficult-to-cut materials, of which the mechanical properties are listed in Table 2. The machine tool used for turning process is the Doosan PUMA 4005 horizontal lathe, of which the key parameters are shown in Table 3. The selected cutting tools are mainly consis of TiAIN-coated cemented carbide turning tools of different geometric dimensions.

During the processing, the data we collect primarily include the workpiece material, process parameters, NC code and machining features, AE sensor data during the machining process, cutting parameters during the actual cutting process, production time, and tool condition and surface roughness data after machining, etc. Among them, the AE signal is acquired using the "R15α" model sensor of the American Physical Acoustics Company with a sampling frequency of 1 MHz. The tool condition data were collected through the industrial microscope LP-950HD. Tool wear is characterized by the maximum value of the tool flank wear (VB_{max}), while the measurement method of tool breakage is analogous to that of tool wear, measuring the maximum values of the width and depth of the breakage. Here, the breakage width refers to the scale of the breakage along the cutting edge direction, and the breakage depth refers to the scale of the breakage perpendicular to the cutting edge direction. According to the previous research, we distinguish the tool chipping and breakage by the maximum length of the breakage in the direction perpendicular to the cutting edge [65,66]. In this paper, the 0.2 mm depth of damage is selected as the criterion to distinguish the fracture edge from the fracture edge. The surface roughness after machining is measured by the Japanese Mitutoyo SJ-210 portable roughness meter. Roughness measurements are taken at five distinct positions on the same machined surface, and the final average value is considered the roughness value of the machining surface. Table 4 lists the technical specifications and measurement settings of the AE sensor and the portable surface roughness profiler, and Table 5 lists the detailed information of the machining configurations.

Table 5
Details of machining configurations.

Item	Notation	Value
Tool supplier	–	{ZCCCT, LST-DJ, Kennamental}
Tool material	M_c	TiAIN-coated cemented carbide
Rake angle of tool	$\gamma(^\circ)$	−6
Principal cutting edge angle	$\kappa_r(^\circ)$	95
Flank angle of tool	$\alpha(^\circ)$	{0, 8}
Cutting edge radius	r_ε(mm)	{0.2,0.4,0.8}
Cutting edge Inclination	$\lambda_s(^\circ)$	{0, −6}
Tool change time	T_{tc} (s)	60
Maximum flank wear	VB_{max} (μm)	Roughing: 600; finishing:300
Allowable breakage size	B (μm)	Roughing: 200; finishing:0
Workpiece material	M_w	{ZTC4, K4169}
Workpiece diameter	D (mm)	{80, 100, 120, 140}
Cutting length	L_{cut} (mm)	{80, 100, 120, 150}
Air-cutting length	L (mm)	180
Coolant condition	–	Water-based
Machining allowances	κ(mm)	{3, 5, 6, 10}
Allowed surface roughness	R_a (μm)	Roughing: 3.2; finishing: 1.6
Cutting speed	v_c (m/min)	[25, 200]
Feed rate	f_r (mm/r)	[0.05, 0.3]
Depth of cut	a_p (mm)	[0.05, 3]

This paper traced turning processing of 36 aero-engine casings in total, which involved 25 types of machining features and utilized 398 cutting edges from three different tool suppliers: ZCCCCT, LST-DJ, and Kennamental. Given the fact that the collection of labeled data at the machining site will interrupt machining and seriously affect production progress, the tool wear values, breakage conditions, and surface roughness of workpiece are recorded only at each tool change or completion of cutting. A total of 425 samples of tool condition and 171 samples of surface roughness were collected. Among these, 224 cutting edges exhibited normal wear, while 51 cutting edges were chipping and 61 cutting edges were breakage.

6.2. Data preprocessing

After acquiring various types of raw data such as AE data, process data, and tool condition data, it is essential to preprocess these data to facilitate effective model training. First, regarding the issue that the high sampling frequency of the raw AE signals significantly increases the computational cost of model training, we segment the AE raw signals using a time window of 1024 and calculate the corresponding statistical features on the segmented data to form a new time series. Herein, we have totally extracted 13 features related to the tool condition, namely: Rise time (RT), Energy (E), Amplitude (A), Average Frequency (AF), Root mean square (RMS), Average signal level (ASL), Peak frequency (PF), Counts to peak (CTP), Reverberation frequency (RF), Initiation frequency (IF), Signal strength (SS), Absolute energy (ABE), Frequency Centroid (FC). The detailed definition of these statistical features can be found in our previous research [67]. Considering that the cut-in and cut-out processes in the cutting process introduce redundant transient effects to the AE signal and interfere with the training of the model, we remove these parts of the signal features. Furthermore, during the actual processing, the interference caused by the continuous rotation of the machine tool spindle and cutting fluid prevents us from promptly identifying the moment of tool breakage to acquire tool breakage samples. Consequently, we employ Short-Time Fourier Transform (STFT) and Dynamic Time Warping (DTW) to detect the moment of tool breakage. Specifically, we first utilize STFT to generate the spectrogram of the raw acoustic emission signal to obtain short-time high-frequency features (The occurrence of tool breakage produces short-time high-frequency components [66,68]). Subsequently, we apply DTW to ascertain the moment of tool breakage by comparing the similarity of time series on either side of the short-time high-frequency points. Finally, we correlate the moment of tool breakage with the corresponding statistical feature values to obtain tool breakage samples. Each sample in the AE dataset is composed of 256 feature data points, and the shape of each sample is 256 × 13. Besides, to ensure that the AE signals can accurately represent the current tool condition, the final segment is designated as the labeled sample, while the other samples from this cutting process are considered unlabeled samples. The above data preprocessing process is shown in Fig. 10.

The data collected in this article comprise multi-source heteroge-

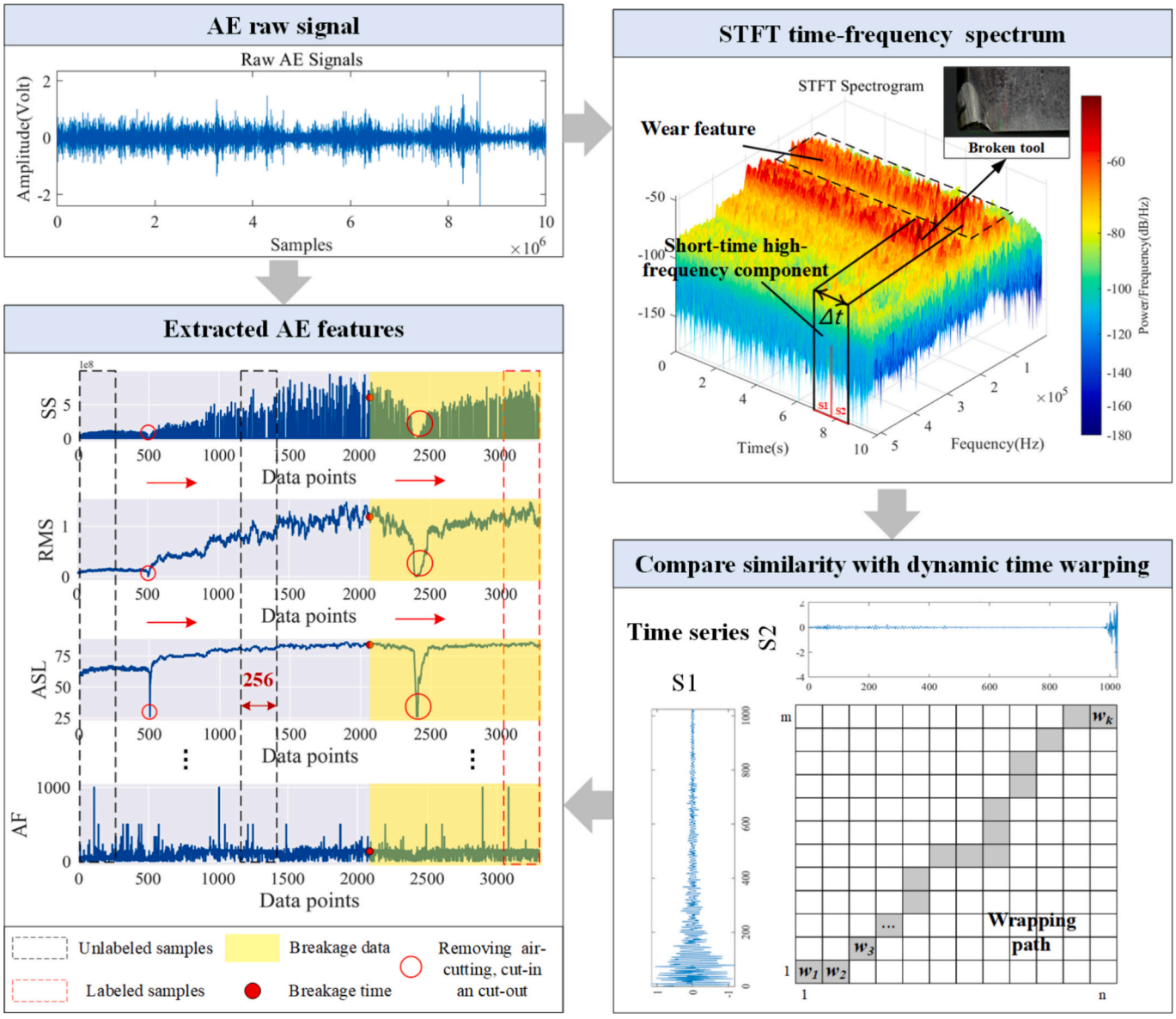

Fig. 10. AE data preprocessing procedure.

neous data such as acoustic emission data, cutting parameters, and tool status, which have different value scales and can affect the effectiveness of model training. Therefore, we apply z-score standardization normalize the 13 statistical features extracted, process parameters, and the state vector involved in agent training along the direction of the input sequence, which can be formulated as:

$$x_i^{scaled} = \frac{x_i - \mu_i}{\sigma_i} \tag{46}$$

Table 6
Hyperparameter settings of MTTS-TCM.

Key hyperparameter	Notation	Optimal value
Epoch	–	T1 = 10, T2 = 300
Batch size	B	16
Learning rate	lr	0.038
Optimizer	–	Adam
Weight coefficient of pseudo-label loss	$\alpha(t)$	α_{max} set to 1
Scaling factor	τ	5
Threshold of CV	k	0.9
Threshold of probability	γ	0.8

where x_i denotes the input sequence, $\mu_i$and σ_i indicates the mean and standard deviation of x_i, respectively.

6.3. Evaluation metrics

To quantitatively evaluate the proposed DRL-TCDDO method, we introduce four common evaluation metrics for multi-objective optimization, namely Hypervolume (HV), Generational Distance (GD), Inverted Generational Distance (IGD), and Spacing (SP). The calculation formulas for the first two are given in Eq. (27) and Eq. (28), while the calculation formulas for IGD and SP are as follows:

$$IGD(PF^*, X) = \frac{\sum_{p^* \in PF^*} \min_{x \in X} \left\| p^* - x \right\|}{|PF^*|} \tag{47}$$

$$SP(X) = \sqrt{\frac{1}{|X| - 1} \sum_{i=1}^{|X|} (\bar{d} - d_i)^2} \tag{48}$$

Table 7
Performance comparison of different methods.

Comparison item	MAE	RMSE	R^2	Accuracy	Precision	Recall	F1-score
STL	103.9865	234.7527	0.6936	0.8353	0.8499	0.8353	0.8068
MTL	103.7869	286.5179	0.5435	0.8706	0.8812	0.8706	0.8518
STTS	81.4924	142.1164	0.9342	**0.9419**	**0.9432**	**0.9419**	**0.9361**
Ours	**47.5154**	**88.8135**	**0.9484**	0.9048	0.9155	0.9048	0.8957

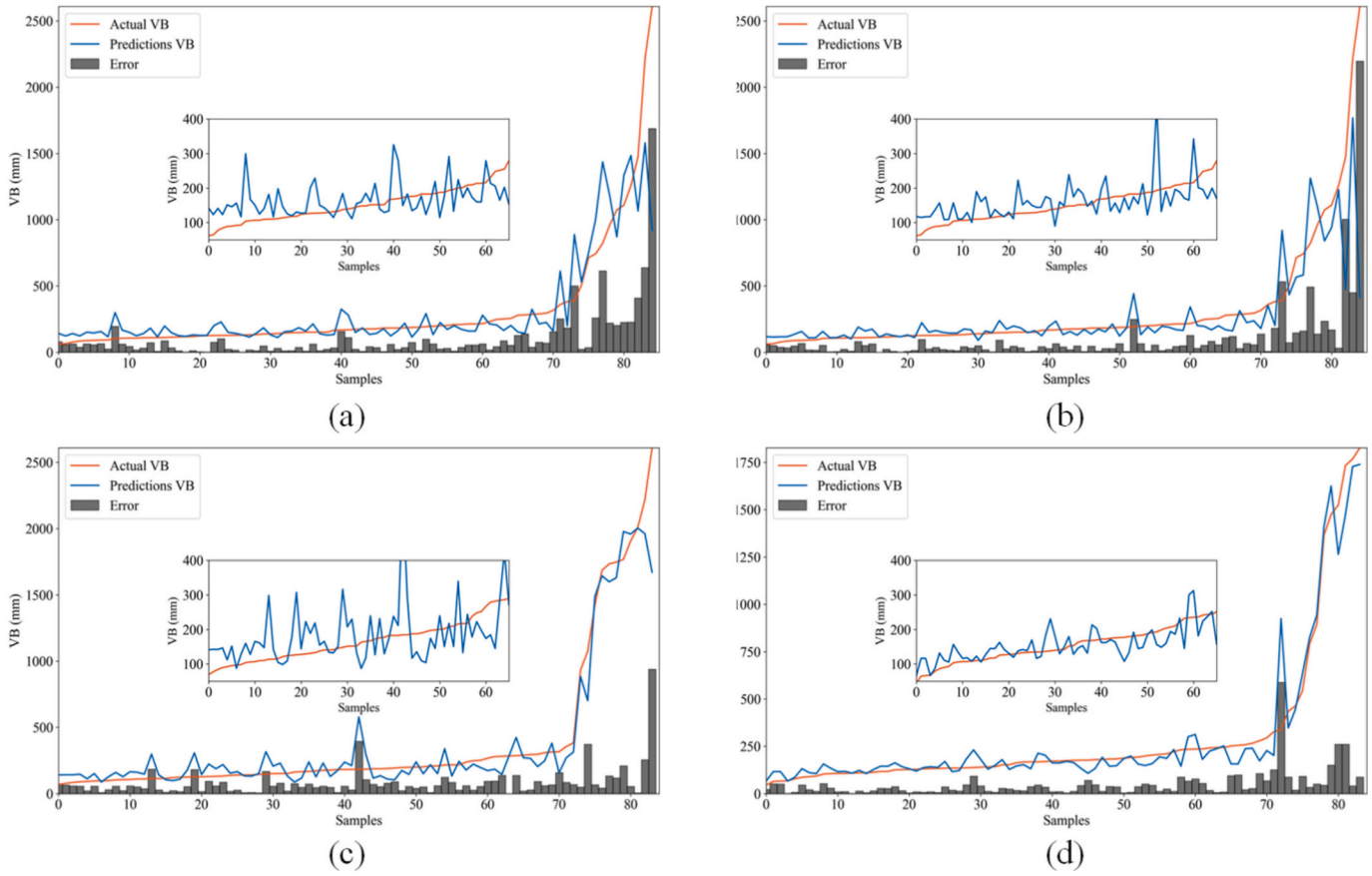

Fig. 11. Predicted VB curves of 4 different methods: (a) STL, (b) MTL, (c) STTS, (d) Ours.

Table 8
Hyperparameter settings of surface roughness estimation model.

Key hyperparameter	Notation	Optimal value
Epoch	–	T1 = 15, T2 = 300
Batch size	B	22
Learning rate	lr	0.017
Optimizer	–	Adam
Weight coefficient of pseudo-label loss	$\alpha(t)$	α_{max} set to 1
Scaling factor	τ	3
Threshold of CV	k	1.1

where X denotes the objective value of the obtained solution set, PF^* is the true Pareto front. $\|\cdot\|$ indicates the Euclidean distance. $|PF^*|$ and $|X|$ denote the number of the true Pareto front and solution set, respectively. $d_i = \min_{j\in X, j\neq i}\|x_i - x_j\|$ is the minimum Euclidean distance from ith solution to other solutions and $\overline{d}$ is the average value of all the d_i. Among these metrics, a larger HV value indicates superior overall performance of the algorithm, while smaller GD, IGD, and SP values indicate better convergence, overall performance, and uniformity of the algorithm, respectively.

Additionally, several common evaluation metrics for classification and regression tasks are applied to evaluate the performance of the proposed MTTS-TCM and surface roughness modelling. For the regression task, we implement root mean square error (RMSE), mean absolute error (MAE), and coefficient of determination (R^2), which are described as:

$$MAE = \frac{1}{m}\sum_{t=1}^{m}|\hat{y}_t - y_t| \tag{49}$$

$$RMSE = \sqrt{\frac{1}{m}\sum_{t=1}^{m}(\hat{y}_t - y_t)^2} \tag{50}$$

$$R^2 = 1 - \frac{\sum_{t=1}^{m}(\hat{y}_t - y_t)^2}{\sum_{t=1}^{m}(y_t - \overline{y})^2} \tag{51}$$

Table 9
Surface roughness estimation results of different methods.

Comparison item	MAE	RMSE	R^2
Empirical model	1.8798	2.1880	−1.4337
Without T-S	0.5201	0.6752	0.7683
Ours	**0.3015**	**0.4010**	**0.9183**

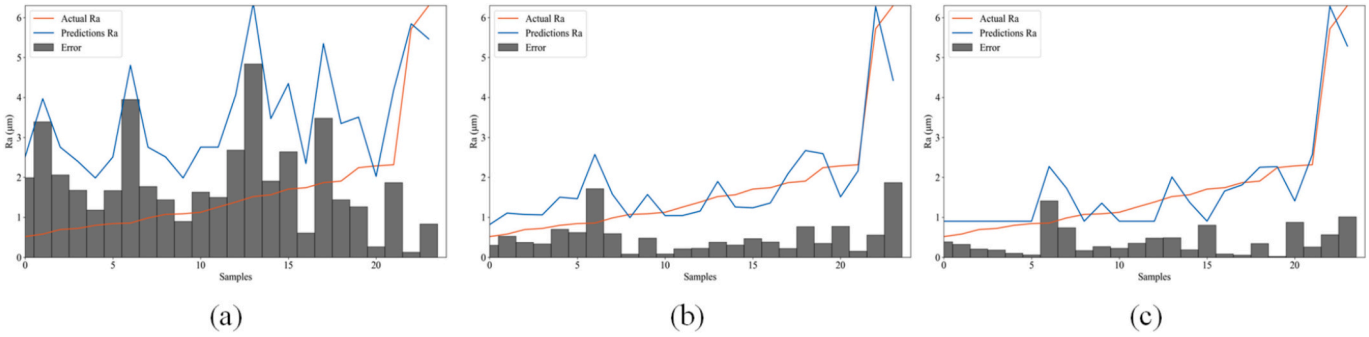

Fig. 12. Predicted R_a curves of 3 different methods: (a) empirical model, (b) without T-S, (c) Ours.

Table 10
Key hyperparameter settings of DRL-TCDDO.

Key hyperparameter	Notation	Value
Training step	T	50,000
Learning rate of actor network	lr_a	0.0003
Learning rate of critic network	lr_c	0.001
Optimizer	–	Adam
Buffer size	M	64
Minibatch size	B	16
Discount factor	γ	0.99
Clip ratio	ε	0.2
Lambda for GAE	λ	0.8
Weight of immediate reward	φ_1	0.1
Weight of sparse reward	φ_2	0.01

where m is the quantity of the test samples, $\widehat{y}_t$ and y_t are the ground truth and predicted values. $\bar{y}$ represent the average ground truth value for all the test samples. For classification task, we select accuracy, precision, recall, and F1 score as the metrics, which are described as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{52}$$

$$Precision = \frac{TP}{TP + FP} \tag{53}$$

$$Recall = \frac{TP}{TP + FN} \tag{54}$$

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \tag{55}$$

where TP denotes the number of positive samples correctly classified as positive; TN represents the number of negative samples correctly classified as negative; FP refers to the number of negative samples mistakenly classified as positive; FN denotes the number of positive samples incorrectly identified as negative samples. In multi-classification tasks, the current class is usually regarded as the positive class, while the other classes are regarded as negative classes.

6.4. Performance evaluation of MTTS-TCM and surface roughness modelling

6.4.1. Applying MTTS-TCM to monitor tool condition

We first randomly sample 20 % of the labeled data as the test set, and utilize the remaining labeled data in conjunction with a substantial volume of unlabeled data as training samples for model training. Subsequently, we employ the Optuna library (https://optuna.org/) to search for the key hyperparameters of MTTS-TCM, which is shown in Table 6. Finally, to eliminate the impact of randomness on model evaluation, we repeat the experiment ten times and take the final average result as the evaluation result of the proposed model.

Table 7 presents the prediction results of MTTS-TCM and lists the prediction results of three comparative methods to highlight the superiority of this approach, which are: traditional single task learning (STL), MTL, and single-task teacher-student architecture (STTS). We emphasize the best metrics in bold font. From the table, it is evident that our method achieves the best results in the tool wear monitoring task, with an R^2 value of 0.9484, which is an improvement of 36.7359 %, 74.4986 %, and 1.52 % compared to the three comparison methods. Although the performance result on the breakage task is not the best, it still achieved a competitive effect, with an F1-score close to 0.9, which demonstrates that MTTS-TCM can achieve the best performance with limited labeled data.

Fig. 11 furthermore gives the predicted VB curves and the corresponding error bars, which can intuitively compare the tool wear prediction results. Here we display the actual VB values for the test sample, arranged in ascending order. As shown in the figure, some cutting edges have a relatively large VB value due to the occurrence of chipping or even breakage during the cutting process. It is important to emphasize that due to the difficult machining characteristics of aircraft engine casings, tool chipping and breakage occur relatively frequently, resulting in generally high VB values, and thus the overall prediction error is relatively large. Tool chipping and breakage are relatively extreme failure mode, with few samples produced and located at the edge of the training data distribution. In addition, their signals exhibit abrupt high-amplitude changes and non-stationary characteristics, resulting in insufficient representation by the model and causing prediction fluctuations. The proposed MTTS-TCM adapts a hard parameter sharing mechanism to enable low-level feature sharing between wear and breakage tasks, combined with pseudo-label learning in a teacher-student framework and a dynamic weighting strategy, to fully leverage unlabeled and limited labeled data to enhance the model's robustness, alleviating this phenomenon to a certain extent.

6.4.2. Surface roughness estimation

Following the procedure outlined in Section 6.4.1, we also randomly selected 20 % of the labeled dataset as test set and utilize the remaining labeled data and all the unlabeled data for model training. The key hyperparameters of the surface roughness estimation model are also determined using Optuna library, which is listed in Table 8. Besides, considering that in the current research on cutting parameter optimization, empirical models are mostly used to model surface roughness, which is usually formulated in Eq. (55). This paper takes the surface roughness estimation method based on the empirical model as the comparative method.

$$R_a = \begin{cases} \dfrac{v_c \times f_r}{\cot k_r + \cot k_r'} & (r_\varepsilon \to 0) \\ \dfrac{(v_c \times f_r)^2}{8 r_\varepsilon} & (r_\varepsilon \neq 0) \end{cases} \tag{56}$$

where k_r and k_r' denote the principal cutting edge angle and auxiliary

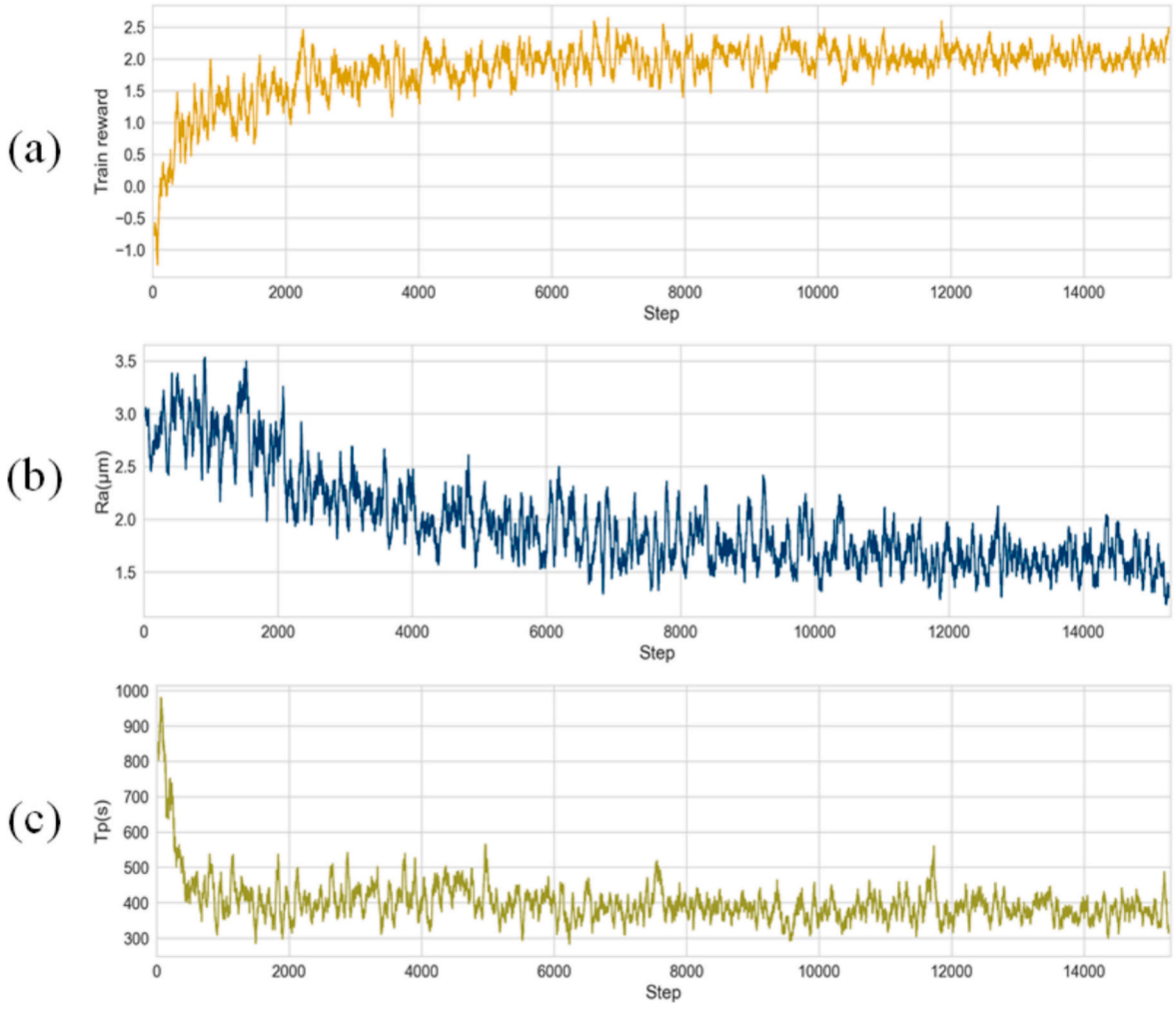

Fig. 13. Convergence curves of DRL-TCDDO during training procedure: (a) rewards, (b) surface roughness, (c) production time.

Table 11
Machining configurations of two optimization cases.

Information	Case1	Case2
Tool supplier	ZCCCT	Kennamental
Tool material	TiAlN-coated cemented carbide	TiAlN-coated cemented carbide
Rake angle	−6	−6
Cutting edge radius	0.8	0.4
Cutting edge Inclination	0	0
Coolant condition	water-based	water-based
Machining allowances	6	10
Cutting length	120	80
Workpiece diameter	80	140
Workpiece material	ZTC4	K4169

cutting edge angle, respectively.r_ε is the cutting edge radius.

Table 9 shows the surface roughness prediction results of our method and two comparative methods, which are empirical model and model without teacher-student architecture (without T-S), and the best metrics are highlighted in bold. And the corresponding predicted curves and errors are depicted in Fig. 12. It can be found from Table 9 and Fig. 12 that the modelling method proposed in this paper has the best effect on surface roughness estimation. Compared with the two comparative methods, the RMSE of our method decreased by 81.6728 % and 40.6102 %, respectively, and the R^2 reached an impressive value of 0.9183. while the empirical model performs the worst, R^2 is even less than 0, we believe that this occurs because the traditional empirical model only considers the relationship between R_a and spindle speed, feed rate, and tool geometry, in which the dynamic change of tool geometry is difficult to measure during machining process. In addition, the tool condition will also affect the surface roughness to a large extent, in which the surface quality will be seriously reduced by chipping or breakage [62,63]. However, the traditional empirical model ignores the influence of tool condition degradation on surface roughness.

6.5. The optimization performance of DRL-TCDDO

6.5.1. The overall performance of our proposed DRL-TCDDO

This section mainly introduces the optimization effect of the proposed cutting parameter optimization method in actual processing tasks. Before performing the cutting parameters optimization, the proposed DRL-TCDDO requires multiple rounds of iterative training to accumulate knowledge for optimizing parameters during the machining. The results in Section 6.4 illustrate that the MTTS-TCM and surface roughness estimation model we proposed can construct an accurate virtual machining environment, providing a good foundation for the interaction between the agent and the environment. The parameter configuration utilized during the training process, and the hyperparameter search results for the proposed method are presented in Table 10.

During the offline training process, the agent interacts with the virtual processing environment to obtain new data for training. An important reference for determining whether the model training has

Table 12
The overall performance of our method in two cases.

Metrics		Case 1		Case 2	
		Without tool condition	Considering tool condition	Without tool condition	Considering tool condition
HV		9311.4906	9417.4641	8935.371	9288.1748
GD		30.6078	15.6169	41.068	42.9303
IGD		9.4009	4.6042	31.7017	3.9362
SP		13.3391	19.8791	11.2004	6.7258
CPU (s)		75.9603	37.9172	114.1750	55.8414
T_p (s)	Min	116.5024	103.3346	167.5079	156.0868
	Average	161.0457	137.6298	189.7691	173.4391
	Max	1328.9013	627.4531	1157.9151	1342.2363
R_a (μm)	Min	0.8206	0.6595	0.6236	0.5122
	Average	1.0828	0.8181	0.9030	0.6352
	Max	5.5031	6.9218	6.9066	7.6387

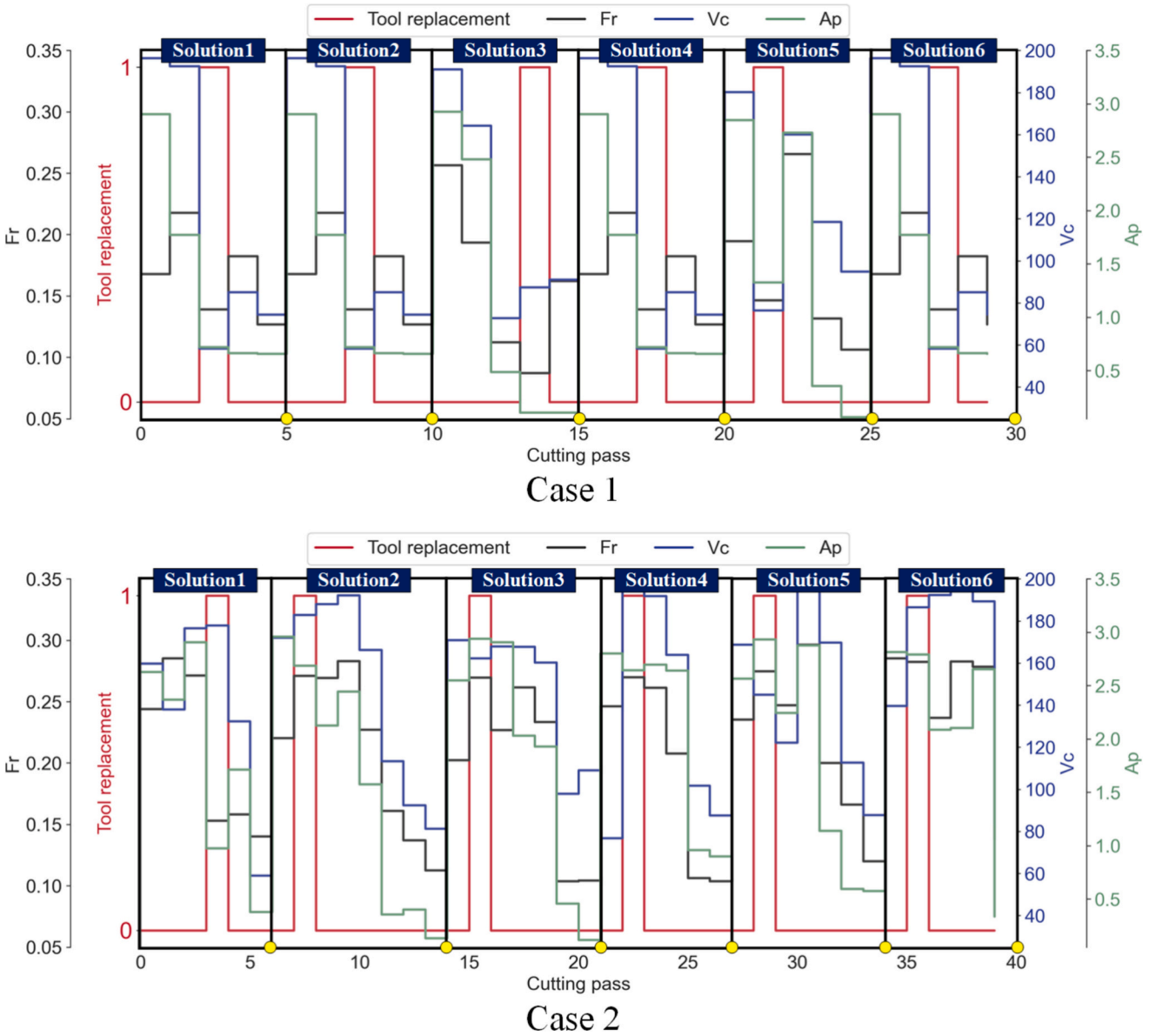

Fig. 14. Visualization of individual solution results in the solution set.

converged is that the reward curve fluctuates within a narrow range. Hence, Fig. 13 gives the trends of changes in rewards, R_a, and T_p during the training procedure. From the figure, it is obvious that as the number of iterations increases, the reward value obtained by DRL-TCDDO from the environment gradually rises, and its fluctuations diminish, approaching a convergent state. Additionally, R_a and T_p also show gradual improvement and optimization throughout the training process, indicating that the agent has accumulated knowledge for optimizing

Table 13
Performance comparison of different optimization algorithms.

Case	Methods	Metrics				
		HV	GD	IGD	SP	CPU(s)
Case 1	MOPSO	8889.0853	51.9005	65.0652	40.7391	6909.524
	SA	7816.8113	32.6892	139.8983	87.2665	9621.4812
	NSGA2	9430.2534	12.1728	20.9723	12.3908	6232.6127
	NSGA3	9376.6729	52.894	30.1463	11.0048	6262.29
	MOEA/D	7634.1968	92.9869	11.1891	41.3173	5890.0232
	AGEMOEA	9307.8589	60.6064	27.1017	10.5982	6351.7766
	CTAEA	8341.7861	135.5658	12.5002	27.2531	5466.3821
	KGB	8421.332	175.8815	238.981	31.2797	14,727.2488
	SMSEMOA	9307.0796	30.0126	16.9864	**9.6167**	6121.7615
	RVEA	7584.7553	**8.9039**	34.2028	18.452	6374.8476
	DRL-TCDDO	**9508.7791**	12.5261	**2.9263**	17.1955	**45.5031**
Case 2	MOPSO	7264.3587	68.091	274.091	198.5946	14,356.3142
	SA	5634.8449	93.7673	635.5773	344.8726	16,431.4396
	NSGA2	8538.0105	91.5857	75.2433	26.0872	10,536.3643
	NSGA3	8599.2632	77.0289	36.2095	18.5244	9732.3636
	MOEA/D	5911.2643	140.2109	32.3166	99.2227	10,390.7287
	AGEMOEA	8605.9992	41.3158	32.6946	28.3013	9688.1324
	CTAEA	6743.8213	139.0921	340.4226	123.8673	1493.9811
	KGB	7002.934	296.6392	275.7915	101.7844	19,975.5507
	SMSEMOA	8707.2338	47.3948	54.8852	16.0324	9941.4129
	RVEA	7101.9699	**8.4583**	28.9266	59.688	9919.4479
	DRL-TCDDO	**9288.1748**	42.9303	**3.9362**	**6.7258**	**55.8814**

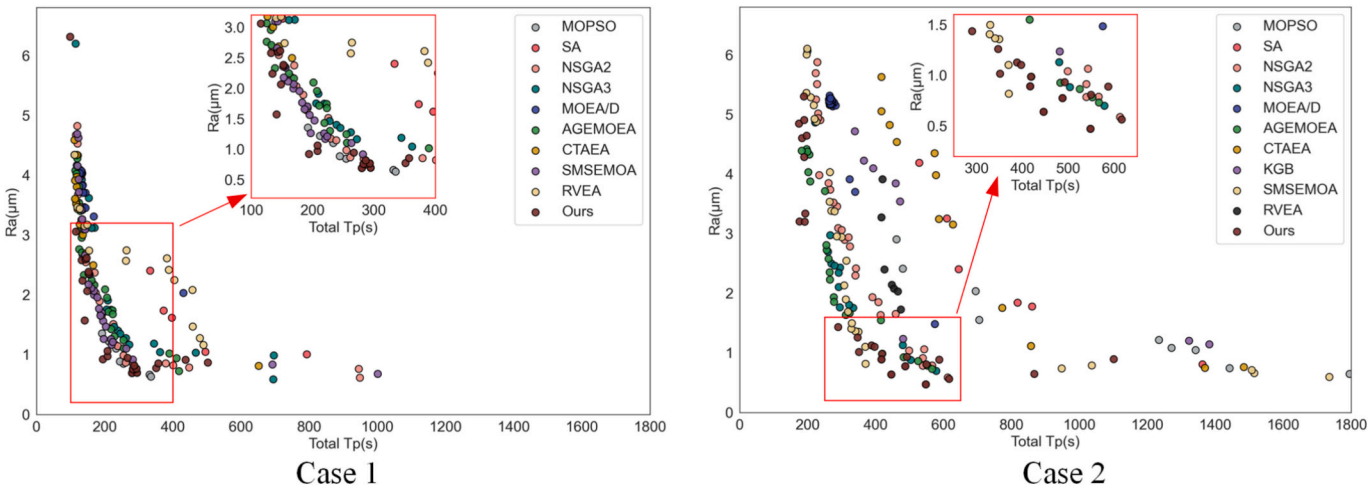

Fig. 15. Approximate Pareto fronts of different algorithms.

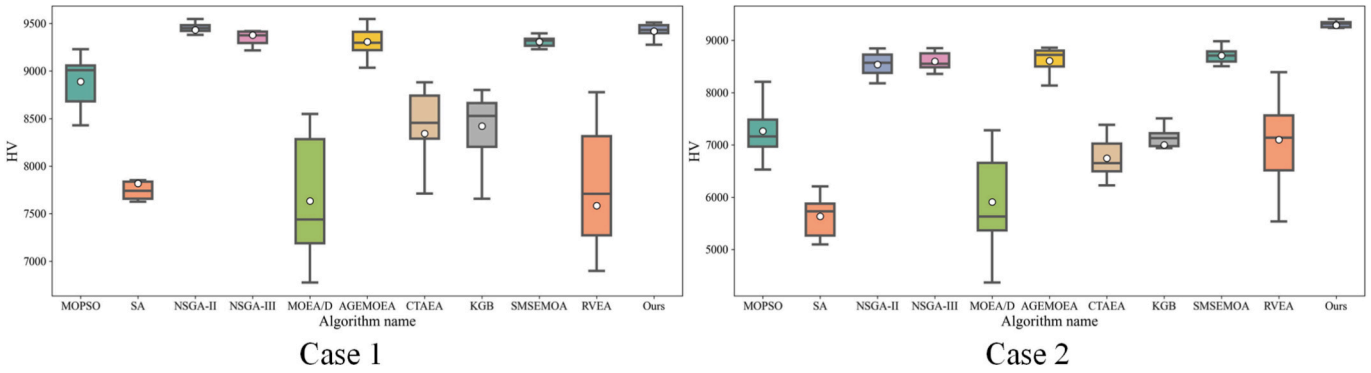

Fig. 16. Box-and-whisker plots of different algorithms on HV metric.

cutting parameters.

In the online optimization stage, the proposed MTTS-TCM first monitors the tool condition after each cutting pass. Then, DRL-TCDDO selects the appropriate actions (changing the tool or adjusting the cutting parameters) based on the tool condition and the current machining status. This paper first conducts online optimization for 2 machining configurations, uses the DOOSAN PUMA 4005 CNC machine tool to process the compressor casing, of which the information is shown

Table 14
The performance of DRL-TCDDO under different machining configurations.

Machining configurations			Metrics				
k	L_{cut}	D	HV	Spacing	GD	IGD	CPU (s)
3	150	80	9812.4019	30.5920	10.3922	8.5356	25.3919
5	120	100	9551.9979	22.4104	3.8647	8.9554	33.6325
6	100	120	9098.9299	18.6597	6.8410	41.3312	67.1670
6	120	80	9417.4641	15.6169	4.6042	19.8791	37.9172
10	80	140	9288.1748	42.9303	3.9362	6.7258	55.8414
Average			9433.7937	26.0419	5.9277	17.0854	43.9900

in Table 11. Besides, to verify that it is necessary to consider the tool condition degradation during the optimization of cutting parameters, we further compared the optimization results without considering the tool condition, which is shown in Table 12. The CPU(s) metric in the table directly reflects the computational efficiency of the algorithm, measuring the time consumed by the algorithm to obtain the corresponding solution set based on the machining states. We can observe from Table 12 that the optimization results considering tool condition degradation perform significantly better in both cases, which suggests that the optimal combinations of cutting parameters frequently vary with the tool condition. To observe more clearly the potential coupling relationship between tool condition and the selection of cutting parameters, this paper describes the changes in the solutions obtained under two cases at each cutting pass (randomly selecting 6 solutions from the solution set and demonstrate them by splicing them according to the cutting pass, where the yellow dot indicates the end of the current solution), which can be depicted in Fig. 14. It can be seen that the well-trained agent tends to select larger cutting parameters at the beginning of the cutting process in order to minimize production time. Subsequently, the agent will choose to implement tool changes, as a combination of large cutting parameters may lead to a rapid tool condition degradation, necessitating replacement to avert catastrophic outcomes. Moreover, to maintain the quality of subsequent finishing, the cutting parameters are typically reduced gradually during the final few cutting passes. Another interesting phenomenon is that, due to excessively frequent tool changes and lower cutting parameter combinations will diminish production efficiency, we can observe that the agent attempts to minimize the number of tool changes as much as possible, and in the last few cutting passes, it does not simply reduce the cutting parameters, but fluctuates within a specific range. This indicates that the agent has grasped the delicate balance between quality and production efficiency, thereby offering valuable insights for selecting cutting parameters in actual manufacturing processes.

6.5.2. *Performance comparison with traditional optimization methods*

To comprehensively evaluate the superiority of the proposed method in this paper, we introduce several widely used algorithms for multi-objective optimization for comparison, which are Multi-Objective Particle Swarm Optimization (MOPSO), Simulated Annealing (SA), Non-dominated Sorting Genetic Algorithm II and III (NSGA-II and NSGA-III), multi-objective evolutionary algorithm Based on decomposition (MOEA/D), Adaptive Geometry Estimation-based MOEA (AGEMOEA), Constrained Two-archive Evolutionary Algorithm (CTAEA), Knee-based Genetic Algorithm (KGB), S-Metric Selection Evolutionary Multi-Objective Algorithm (SMSEMOA), and Reference Vector Guided Evolutionary Algorithm (RVEA). To ensure fairness, all comparative

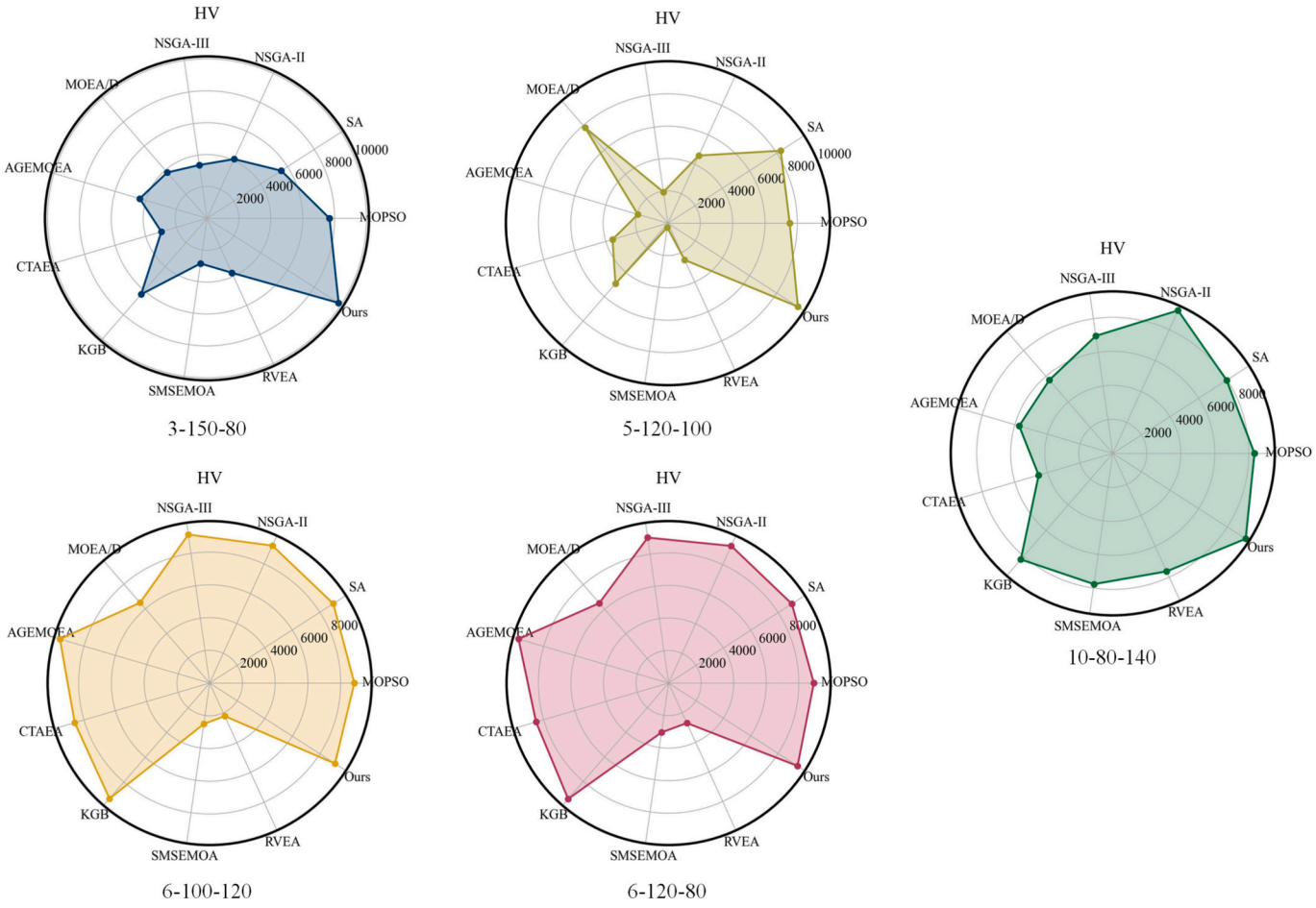

Fig. 17. HV values of different algorithms on each machining configuration.

Table 15
The main change of new machining configurations.

Item	Notation	Value
Machine tool	–	Spinner TC-800 {X/Y/Z axis travel range: 360/120/900 mm; Rapid feed of X/Z axis: 18/24 m/min; Maximum spindle speed: 4000 rpm; Maximum turning length: 900 mm; Maximum turning diameter: 500 mm}
Tool supplier	–	{ZCCCT, GESAC, SANDVIK, KORLOY, Kennamental}
Cutting edge radius	r_ε(mm)	{0.2, 0.4, 0.65, 0.8}
Tool change time	T_{tc} (s)	100
Workpiece diameter	D (mm)	{165, 207, 237, 300}
Cutting length	L_{cut} (mm)	{50, 80, 100, 200, 250}
Air-cutting length	L (mm)	360
Machining allowances	κ(mm)	{2, 4, 6, 8, 12}
Allowed surface roughness	R_a (μm)	Roughing: 3.2; finishing: 1.6
Cutting speed	v_c (m/min)	[20, 250]
Feed rate	f_r (mm/r)	[0.05, 0.35]
Depth of cut	a_p (mm)	[0.02, 2.5]

Table 16
The performance of DRL-TCDDO under entirely new and unseen machining configurations.

Machining configurations			Metrics				
k	L_{cut}	D	HV	Spacing	GD	IGD	CPU (s)
2	250	165	8869.9421	6.5633	2.1044	6.1905	11.2331
4	200	207	9024.8429	14.5673	5.2967	18.2675	33.6325
6	100	237	9141.5289	9.7248	7.4562	19.7141	44.1466
8	80	300	9005.1355	20.4469	7.4419	23.9262	47.5346
12	50	300	8669.9421	7.5745	10.9887	20.3639	80.7776
Average			8942.2783	11.7754	6.6576	17.6924	43.4649

algorithms are run ten times, and the average values are taken as the final results, where the reference point of the HV indicator is (1800, 6.3). We also use the machining configurations of the above two cases as examples for testing, the results are shown in Table 13. It reveals from the table that our method performs the best overall and can obtain a non-dominated solution set with higher performance within the target space, of which the HV index in the two cases increased by an average of 11.2009 % and 28.2361 % respectively compared with the 10 comparison algorithms. In terms of computational efficiency, DRL-TCDDO far surpasses other comparison algorithms, with an average increase of 99.3324 % and 99.1906 % in CPU metric in the two cases, respectively. This is because DRL-TCDDO transforms the cutting parameter optimization problem into a sequential decision-making process, reduces the spatial dimension of the decision variables. Furthermore, DRL-TCDDO has accumulated a wealth of knowledge on parameter optimization during the training process, enabling the agent to promptly provide optimized results based on its current machining state.

To demonstrate the superiority and stability of our method more intuitively, This paper visualizes the approximate Pareto fronts obtained by each algorithm in the two cases and plots the corresponding box-and-whisker plots of HV metrics after multiple runs, which are shown in Fig. 15 and Fig. 16. As can be seen from Fig. 15 the multi-objective optimization results of different algorithms all converge to the lower left corner, indicating that the virtual machining environment established in this paper can be well combined with each algorithm to solve the corresponding optimization problems. Herein, the solution set obtained by the proposed method is overall closer to the bottom left corner and has a uniform and broad distribution, which indicates that it is closer to the true Pareto front. The results in Fig. 16 shows that the DRL-TCDDO achieves the best HV metric in both two cases with narrowest variation range, which suggests that it demonstrates superior performance and greater stability in the search for optimal values. Besides, this advantage is even more pronounced under conditions with large machining allowances.

6.5.3. Verification of model generalization

The aforementioned DRL-TCDDO requires prior pre-training to accumulate optimized knowledge of cutting parameters for online optimization. This means that model optimization is not conducted for specific processing configurations like metaheuristic algorithms, but rather guides the agent to learn long-term strategies through the reward function, enabling it to exhibit good generalization performance across different turning operation conditions. However, there are inevitable differences between the virtual machining environment and the real machining environment. For instance, the virtual environment omits the dynamic characteristics of the machine tool, the thermal-mechanical information of the contact between the tool and the workpiece, and the influence of the coolant, etc. Thus, this paper first employs trained agents directly to optimize testing tasks of different configurations for turning operations, with primary variations in machining allowance, cutting length, and workpiece diameter, of which the test results are presented in Table 14. It can be seen that the method proposed in this paper can achieve competitive optimization effects under five different processing configurations, which proves the generalization and universality of our method. Furthermore, we compared the performance of DRL-TCDDO and the traditional optimization algorithm mentioned above under different machining configurations. To clearly demonstrate the superiority of our method, we visualize the HV results of different algorithms under each working condition in the form of radar charts, which is illustrated in Fig. 17. Due to the blind and random in the optimization process of metaheuristic algorithms, these methods do not effectively leverage the knowledge of the manufacturing process. Consequently, they tend to exhibit suboptimal performance and limited stability across varying working conditions, which necessitates re-solving and incurs substantial time and computational resource expenditures. While the proposed method has accumulated a large amount of manufacturing process knowledge under different turning conditions during the training process. It can make decisions directly in scenarios where the machining state changes rapidly without the need for a large number of search operations.

To further enhance the persuasiveness of this research, we conducted another aero-engine casing processing experiment with brand-new machine tool, workpiece and cutting tool, and collected the corresponding data to verify the generalization of the DRL-TCDDO. The main change in the new processing experiment configuration are listed in Table 15. The methods for collecting AE signals and measuring surface, as well as the corresponding data preprocessing, remain unchanged. Finally, we collected the processing data of three different aero-engine casings, which included a total of 27 parts and eight processes. During this process, we traced the condition of 193 cutting edges, and recorded a total of 200 samples of tool condition and 169 samples of surface roughness. Among them, the samples of normal tool wear were 130, and the samples of chipping and breakage accounted for 13.5 % and 21.5 % respectively. Similarly, we have listed the results of testing DRL-TCDDO under completely new and unseen working conditions, as shown in Table 16. From the results in the table, it can be seen that under completely new test conditions, DRL-TCDDO still maintains good overall performance. Only the HV and GD metrics showed a decline, with decreases of just 5.2102 % and 12.3134 % respectively.

7. Conclusion and future work

Efficiency and quality have always been the goals pursued by the manufacturing industry, and optimizing process parameters, as an effective means to achieve efficient and high-quality production, how to

select suitable cutting parameters to improve the machining process has become a research hotspot, attracting widespread attention in both academic research and industrial applications. However, previous research has either neglected the dynamic changes in tool conditions throughout the cutting process or assumed a static process optimization with fixed tool condition. To address this gap, this paper proposes a dynamical cutting parameters optimization method that comprehensively considers the cutting tool condition degradation, transforming the optimization problem into MDP and solving it based on RL method: Proximal Policy Optimization (PPO), which enables rapid and accurate decision-making under various machining configurations. Based on the findings of the case study, we can draw the following conclusions:

- To achieve an effective combination of TCM and dynamic decision-making, this paper presents a multi-task teacher-student architecture for TCM (MTTS-TCM), which accurately monitors tool wear and breakage condition using limited labeled data from the manufacturing site, providing a basis for subsequent cutting parameter optimization. The proposed model achieves an R2 of 0.9484 for tool wear value monitoring, and an F1 score of 0.8957 for breakage condition monitoring.
- Furthermore, we apply the aforementioned teacher-student architecture to construct a high-precision and universal objective function, namely surface roughness modelling. This does not require additional professional knowledge or understanding of the underlying mechanism, making it more cost-effective. The R^2 of the surface roughness monitoring of the proposed modelling method reaches 0.9183.
- We transform the dynamic optimization problem of cutting parameters for multiple turning operations into a sequential decision-making process, constructing a corresponding virtual machining environment for agent interaction, and effectively learn optimization knowledge covering various turning processes based on the PPO algorithm. Compared to the optimization results that do not consider tool condition degradation, our method achieves improvements in HV, GD, and IGD of 1.1381 %, 48.9774 %, 51.0238 % and 3.9484 %, 4.5347 %, 87.5836 %, respectively in two cases. Compared with 10 multi-objective optimization algorithms, the HV values of the solution set obtained by our method improved by 11.2009 % and 28.2361 % in two cases respectively.
- DRL-TCDDO demonstrates good generalization for turning operations with different machining configurations. It outperforms the comparison algorithms under five different operating conditions, and the average computation time required for decision-making is reduced by over 99 %. In addition, under entirely new and unseen operating conditions, the proposed method demonstrates robustness, with the HV and GD metrics decreasing by only 5.2102 % and 12.3134 %, respectively.

The cutting parameter optimization method proposed in this article is mainly focused on turning operations, and it can be considered for extension to milling, drilling, and other processes in the future. Moreover, the optimization of process parameters in this study is limited to specific manufacturing steps pertaining to certain components, without taking into account the entire processing flow. Consequently, future research may focus on the integration of workshop process route planning and cutting parameter optimization.

CRediT authorship contribution statement

Xuandong Mo: Writing – original draft, Software, Methodology, Conceptualization. **Mingyuan Xia:** Writing – original draft, Conceptualization. **Shenping Mei:** Writing – original draft, Conceptualization. **Xiaofeng Hu:** Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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